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Applied Statistical Causal Inference in Requirements Engineering

A Gentle Introduction to the Analysis of Quantitative Data from Observational Studies

1-5 September 2025
Valencia, Spain

Monday, 1st of September, 14:00-17:30 in Room 2.3

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Statistical Causal Inference

a gentle introduction to advanced analysis of quantitative data

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JulianFrattini/**bda4sci** Public

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Meet your Instructors

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Overview

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Session	Time	Content	Style
Introduction	14:00-14:15	State-of-the-art, motivation, and goals	Lecture
Fundamentals & Notation	14:15-14:45	Fundamental terminology in statistics, and notation used throughout this tutorial	Lecture
Causal Inference I	14:45-15:30	Mediation and confounders	Lecture
Break	15:30-16:00		
Causal Inference II	16:00-16:45	Colliders and model comparison	Lecture & Exercises
Application	16:45-17:00	Statistical causal inference for requirements engineering	Lecture
Conclusion	17:00-17:15	Scientific workflow for causal inference, structured reading list	Lecture
Feedback	17:15-17:30	Reflection, discussion, and closing	Open discussion

Introduction

Goals and Expectations

State-of-the-art

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We would like to
we do not claim cau
on understanding the
tones expressed by the
issue tracking systems
measures. Overall, our
picture of the relations
issue discussions have
influenced.

Sanei, A., Cheng, J., & Adams, B. (20
generated issue discussions. In *202*
Human Aspects of Software Enginee
10.1109/CHASE52884.2021.00009

1 Introduction

Software requirements continue to evolve during software development and maintenance, and the associated risk to cost, project schedule, and quality appeals to the need for increased understanding of the phenomena. The case studies presented in this paper are part of a family of collaborative empirical initiatives, each of which makes progress towards the ultimate goal of requirements change anticipation. Loconsole [1] provides a model to predict requirements change which utilises the observed correlation between the size of a requirements use case document and volatility. While correlation is significant, there is no claim made with respect to causality. Our approach is to use case studies to increase our understanding of the causes and effects of requirements change in order to provide an empirically founded theory upon which to base causal models. To this end, it is necessary to firstly determine an

McGee, S., & Greer, D. (2012). Towards an understanding of the causes and effects of software requirements change: two case studies. *Requirements Engineering*, *17*, 133-155. DOI: 10.1007/s00766-012-0149-0
Referring to Loconsole, A., & Borstler, J. (2005, December). An industrial case study on requirements volatility measures. In *12th Asia-Pacific Software Engineering Conference (APSEC'05)* (pp. 8-pp). IEEE. DOI: 10.1109/APSEC.2005.38

5.5 Threats to validity

External validity The generalizability of results cannot be claimed based on the exploratory case study on one data set. Further data sets are necessary to be investigated and compared to compensate for context factors like the size and domain of the company, the utilized development process, techniques employed in the requirements engineering phase, as well as applied technologies. However, the analyzed data-set represents five years of product development (between 2010 and 2015) with 41 products and 36 software releases. Therefore, the heterogeneity of authors and editors of requirements is high.

Internal validity To ensure that the correlation between the occurrence of causality and the lead-time of requirements is indeed causal and not confounded, further qualitative analysis beyond the data recorded in the respective data set must be performed. The impact of causality on comprehensibility in contrast to other factors of ambiguity has to be addressed in future studies of qualitative nature. Apart from

Frattini, J., Fischbach, J., Mendez, D., Unterkalmsteiner, M., Vogelsang, A., & Wnuk, K. (2023). Causality in requirements artifacts: prevalence, detection, and impact. *Requirements Engineering*, *28*(1), 49-74. DOI: 10.1007/s00766-022-00371-x

Correlation versus Causality

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Correlation:
when we *observe* X,
we also observe Y

Causality:
when we *do* X,
we observe Y.

If we want to make a **positive impact** on the target audience of our research (i.e., RE/SE practitioners), we need to provide **reliable recommendations on how to act** – which we can only derive from **causal** relationships, not correlations.

> We cannot infer causality from data alone – we must also know the process that created that data.[1]

[1]we'll revisit this statement later

Experimental vs. Observational Studies

One ”straightforward” way to distinguish correlation from causality: **conduct a controlled experiment**, where we exert control over a treatment and isolate its effect from all confounding.

Expensive
Recruiting eligible participants is difficult

Unrealistic
Experiments perturb the natural context

Impossible
Some factors are impossible to control

Assumed
The validity of experimental results depends on several conditions

Goals of this Tutorial

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- Convincing you of the **value of statistical causal inference**
- Learning the fundamentals of **drawing causal conclusions** from quantitative data collected in **observational studies**

Learning Outcomes

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Causal Modelling
Communicating transparent causal assumptions in graphical models

Statistical Causal Inference (SCI)
Drawing reliable conclusions from quantitative, observational data

Causal Workflow
Structured process for performing statistical causal inference in an empirical study

Fundamentals & Notation

Syntax and Pedagogics of this Tutorial

Basic Terminology

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"We study the effect of different **document reading techniques** on the **number of identified defects** in **Bachelor- and Master-level** students."

- **Factors**: variables that associate measurements with a construct (e.g., document reading technique, identified defects, student level)
- **Treatment/Exposure**: main independent factor of interest
- **Outcome/Response**: main dependent factor of interest
- **Phenomenon**: main causal effect of interest (**treatment** → **outcome**)

Causal Modeling

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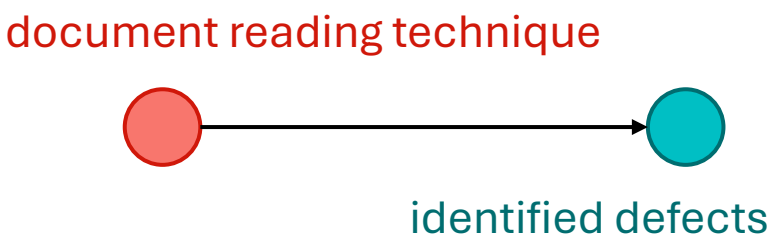
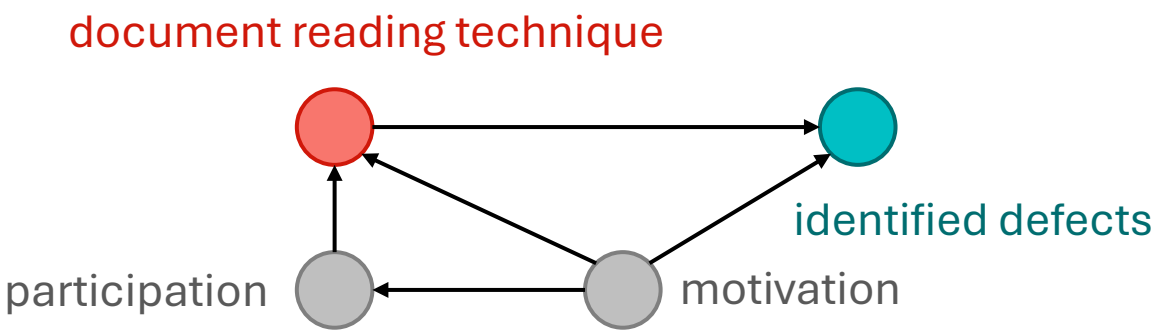
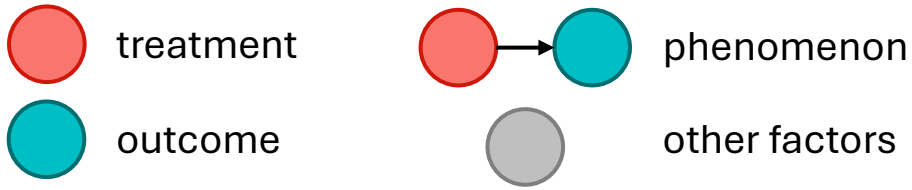
Directed, acyclic graphs (DAGs) can represent causal assumptions, where *nodes represent variables* and *edges represent assumed/potential causal relationships* between those variables.







> DAGs are subjective and represent the view of a domain expert.

Simulations

To demonstrate the process of SCI, we **simulate** data sets. This gives us control over phenomena, i.e., a reliable ”ground truth.”

Discrete uniform distribution

Simulation of an independent variable

Simulation

$$technique = unif\{0, 1\}$$

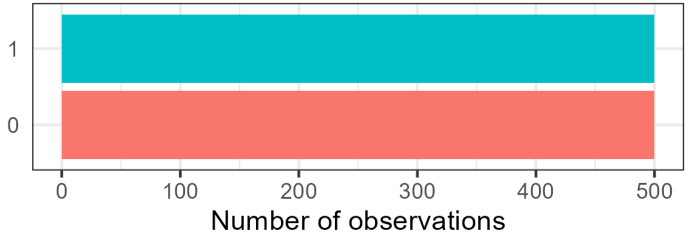
$$defects = Poisson(10 + technique * 5)$$

Simulation of a dependent variable

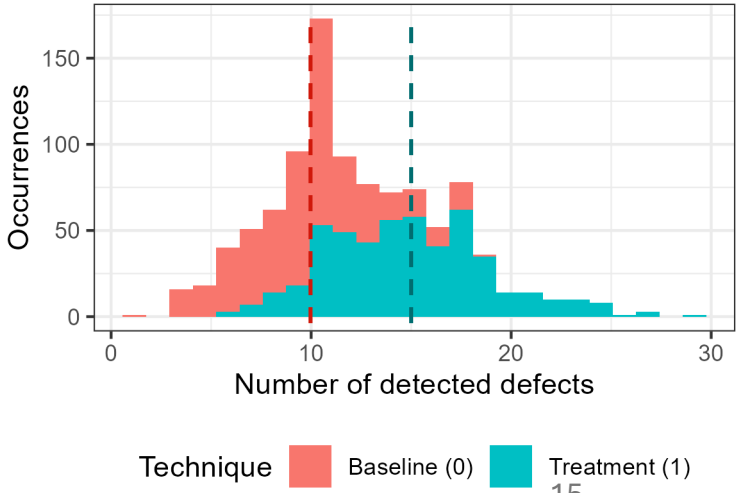
Poisson distribution

Dependency of defects on technique

	technique	defects
1	0	12
2	1	20
3	0	9
4	1	7
5	0	8
6	1	24
7	0	6



$$10 + \quad technique * 5$$



Causal Modeling and Simulations

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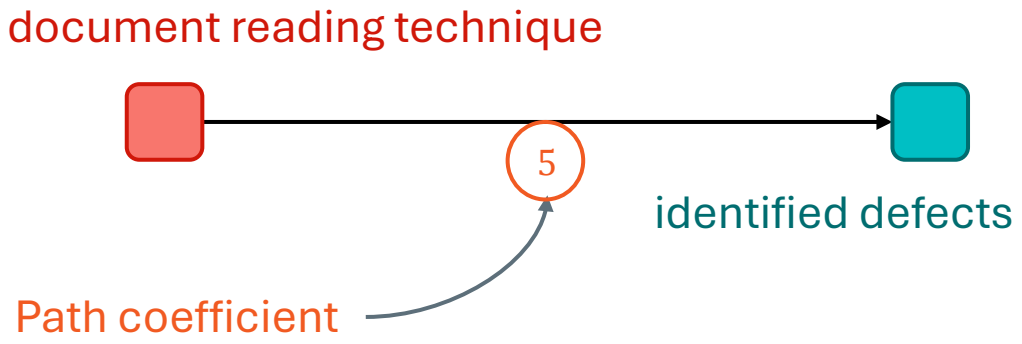
DAGs are typically used to formalize *assumed* causal relationships. They can also be used to represent **simulated** causal relationships.

Simulation

$$technique = unif\{0, 1\}$$

$$defects = Poisson(10 + technique * 5)$$



> Changing the cause by one unit causes the effect to change by <path coefficient> units. It is, therefore, our variable of interest representing effect strength.

Data Analysis Tool: (Linear) Modeling

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+ more accessible — more difficult to use and apply -

- less granular — more granular +

Null-Hypothesis Significance Test

e.g., Student's t-test, Mann-Whitney U Test

(Generalized) Linear (Mixed) Model

outcome — predictor

$$leadtime \sim technique$$

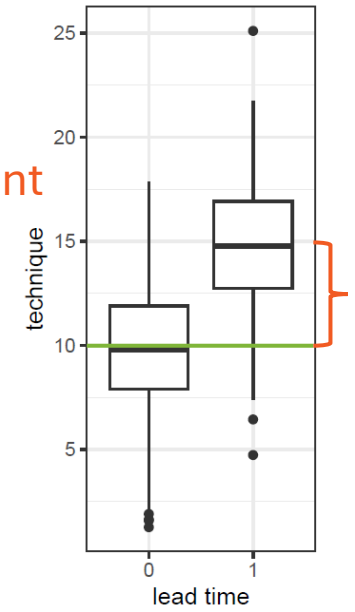
$$leadtime \sim \alpha + \beta_t * technique$$

intercept — path coefficient

```
Call:
lm(formula = leadtime ~ technique, data = d)

Coefficients:
(Intercept)    technique
      9.823        5.032
```

Bayesian Data Analysis

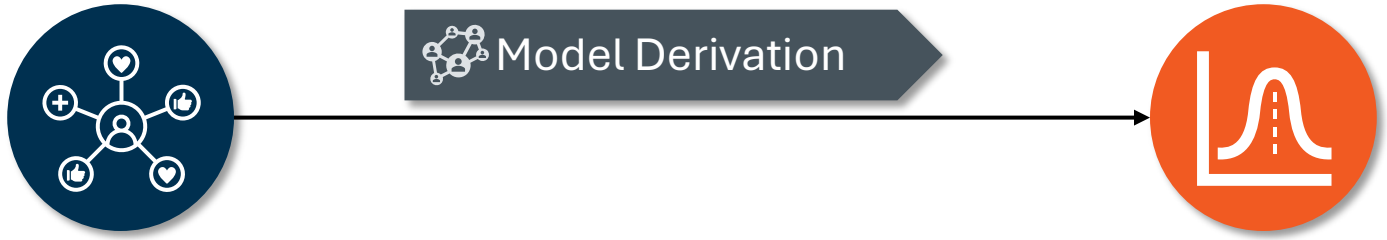


Simulation

$$technique = unif\{0, 1\}$$

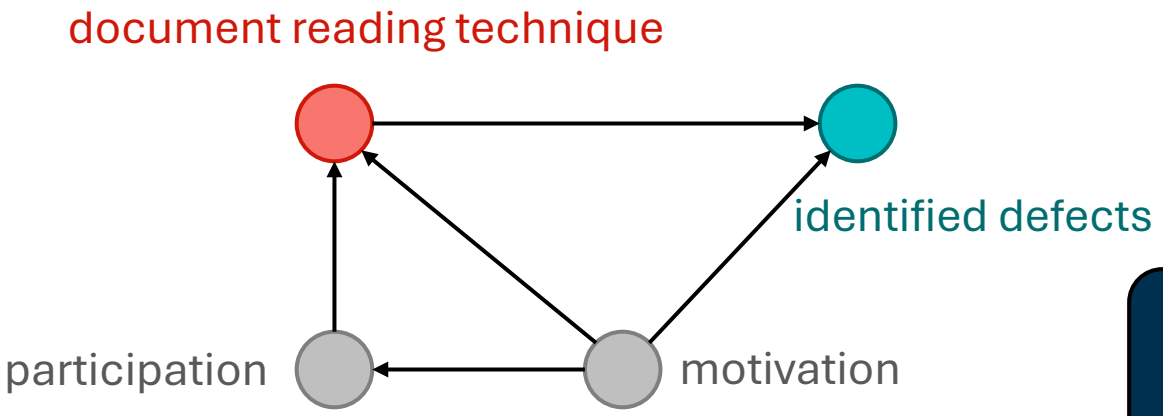
$$leadtime = N(10 + technique * 5, 3)$$

From causal to statistical Models



Causal Models
encode causal assumptions

Statistical Models
implement inference



$$defects \sim technique + motivation$$

> How to derive statistical from causal models is core to statistical causal inference.

Pedagogy of this Tutorial

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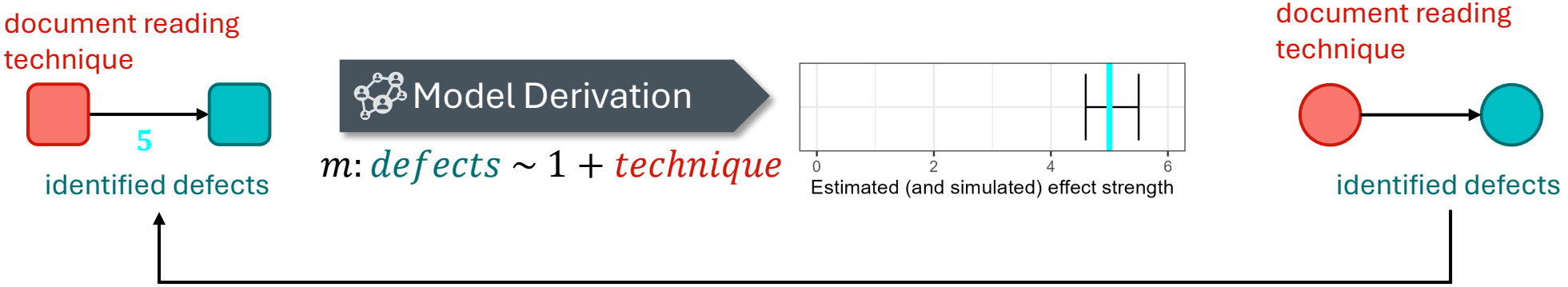
1. Ground Truth
Simulating data based on defined causal relationships

2. Regression
Deriving statistical models and running regression analyses

3. Comparison
Comparing the results of the analysis with the ground truth

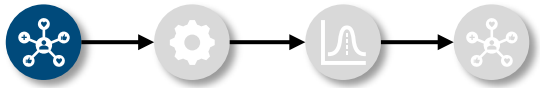
4. Inference
Deducing obtainable knowledge from the analysis results



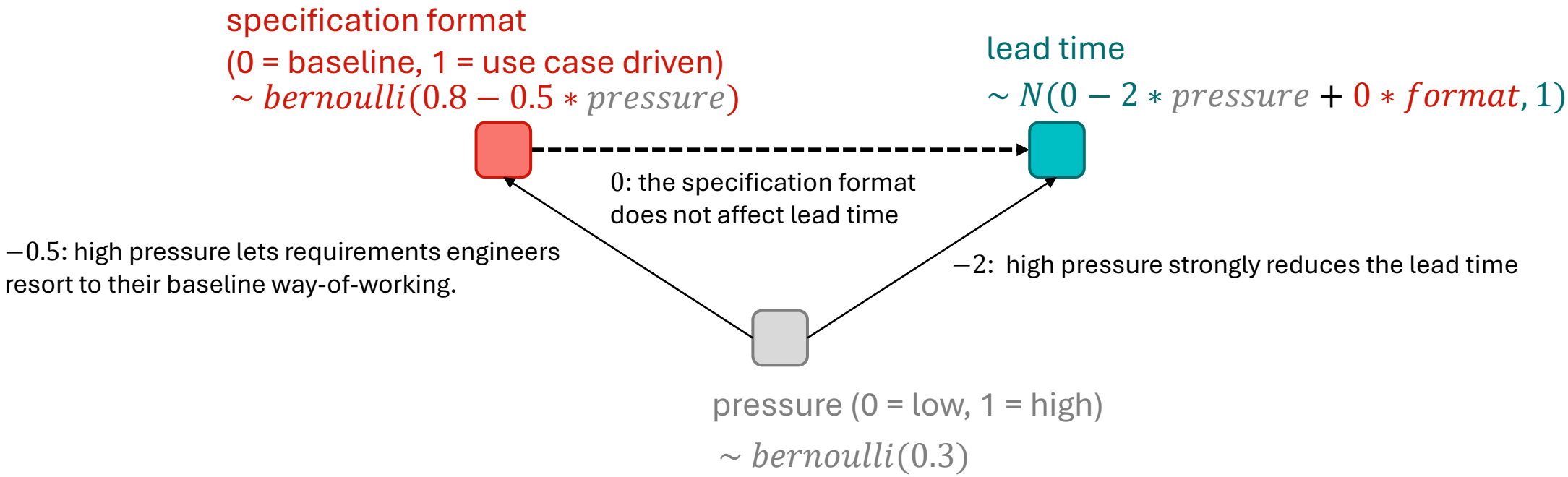
Causal Inference I

Confounders and Mediators

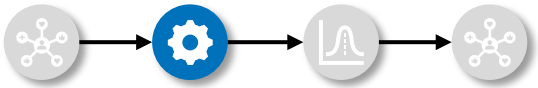
Confounding: Common Causes



"The company pushed for a use case driven requirements specification approach. We investigate whether the **specification format** of a requirement decreased the normalized **lead time**."



Confounding: Common Causes



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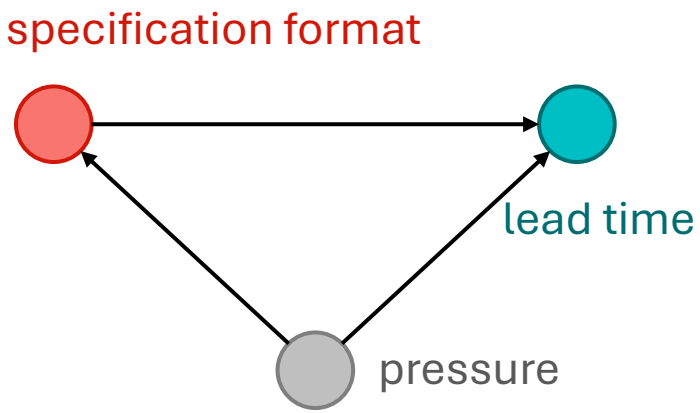
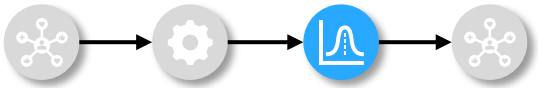


Model Derivation

m_1: $leadtime \sim 1 + format$

m_2: $leadtime \sim 1 + format + pressure$

Confounding: Common Causes



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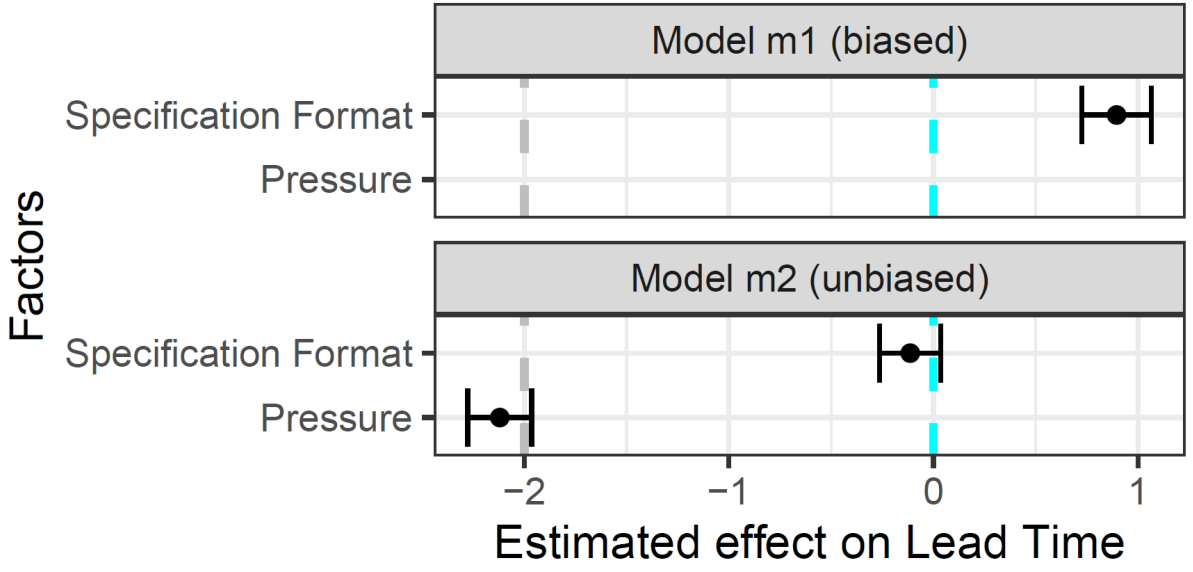

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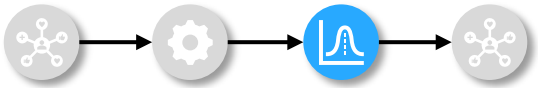
m_2: $leadtime \sim 1 + format + pressure$

Confounding: Common Causes



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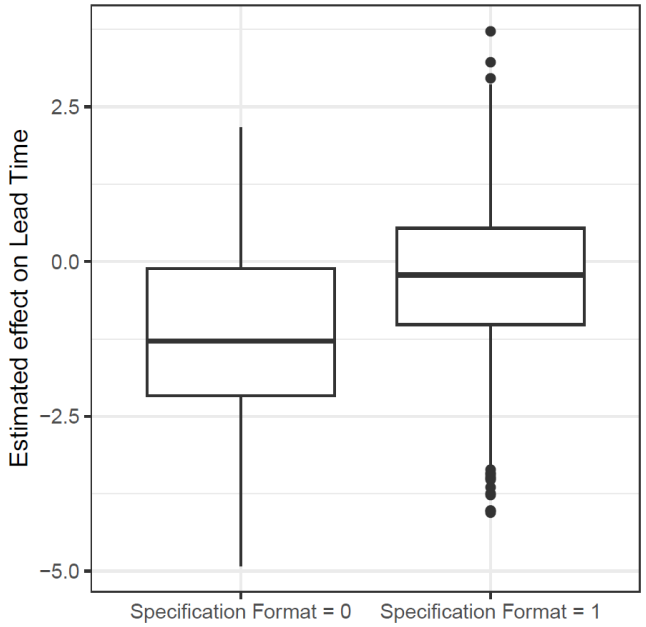

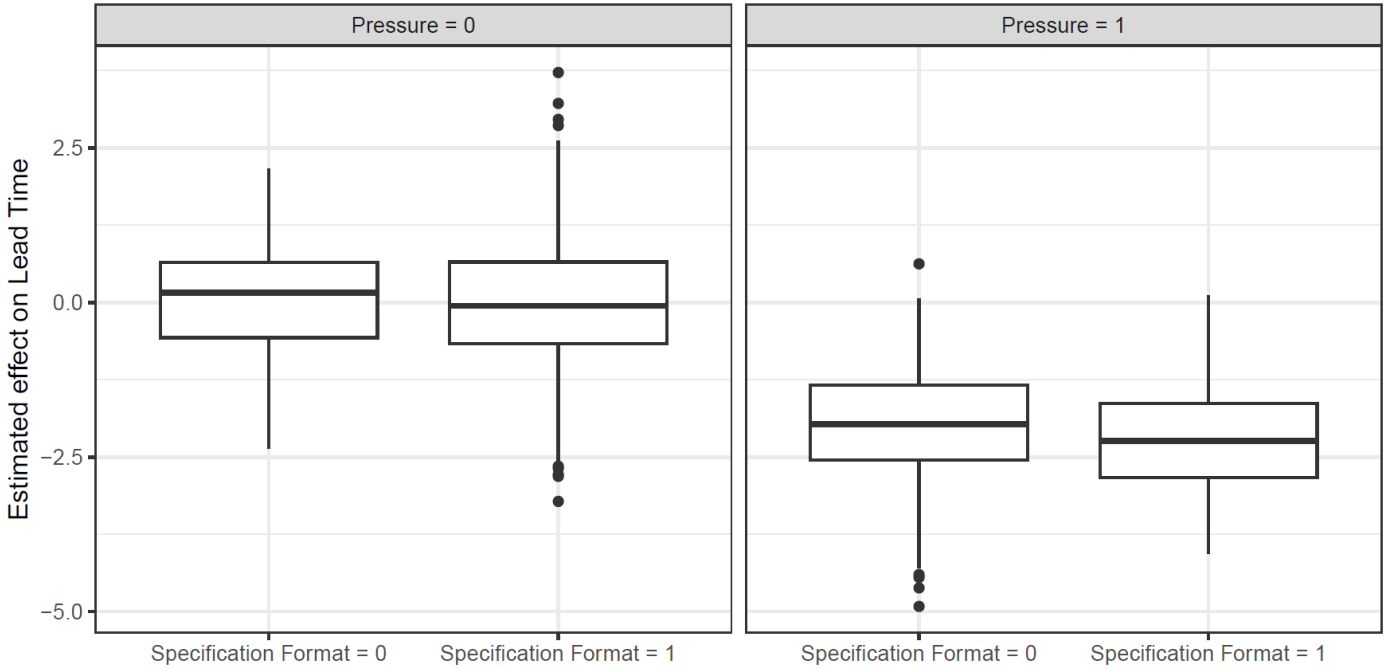

Confounding: Common Causes

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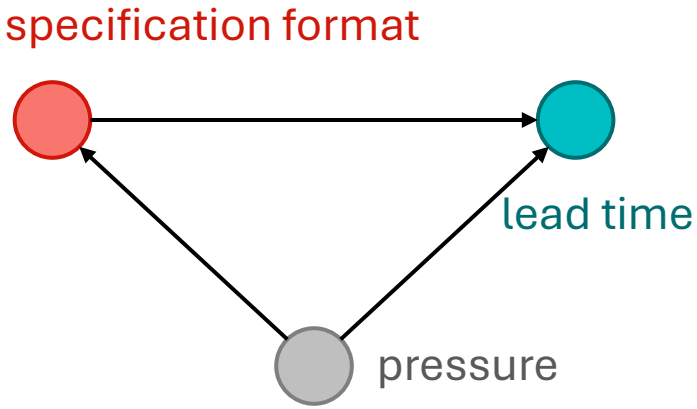


Model Derivation

$\cancel{m_1: leadtime \sim 1 + format}$

$m_2: leadtime \sim 1 + format + pressure$

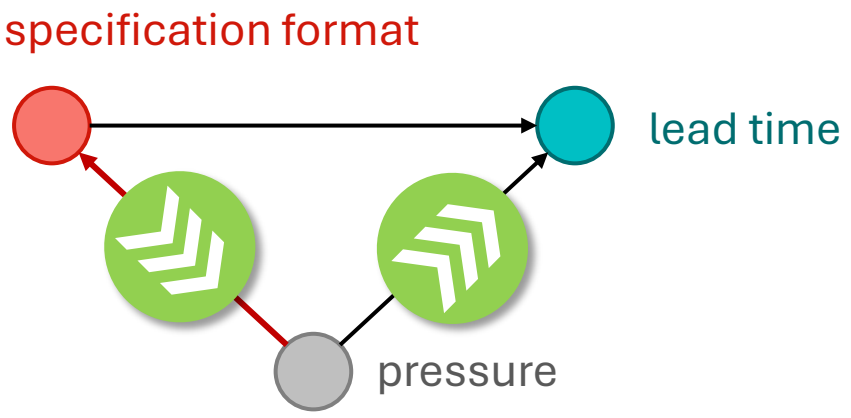
> If both treatment and outcome have a common cause, we need to statistically adjust for (i.e., control) the confounder to de-bias the effect of interest.

Confounding: Common Causes

Whenever specification format = 0, the likelihood that pressure = 1 is much higher which causes a lower lead time.

> "Whenever specification format = 0, the likelihood that pressure = 1 is much higher" is an association, but not causal in this direction!

Remember: observing ≠ doing



When *not* controlling the confounder, "information flows" from the cause through the confounder to the effect in a non-causal direction.

Practical Example of Confounding

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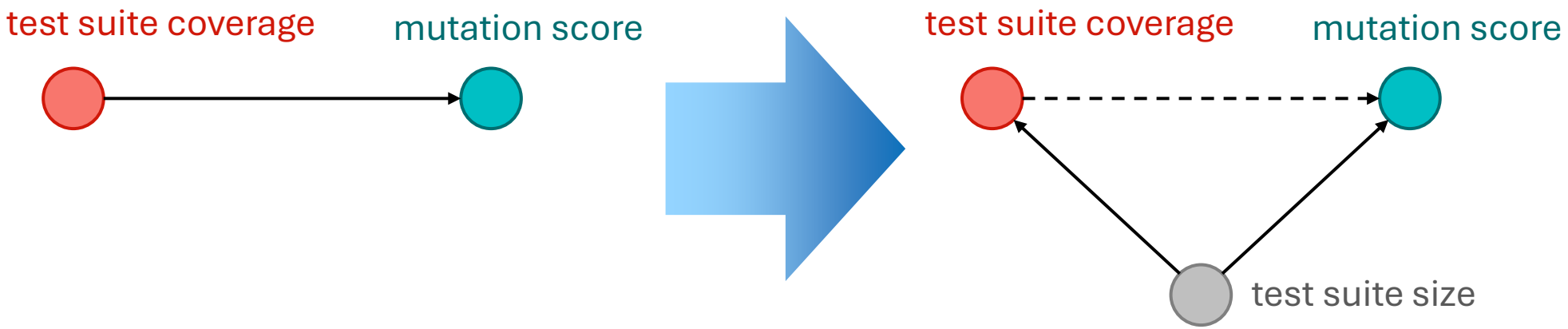
“Coverage Is Not Strongly Correlated with Test Suite Effectiveness”

We found that there is a low to moderate correlation between coverage and effectiveness when the number of test cases in the suite is controlled for. In addition, we found that

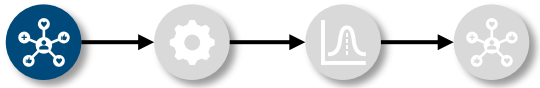


http://dx.doi.org/10.1145/2568225.2568271

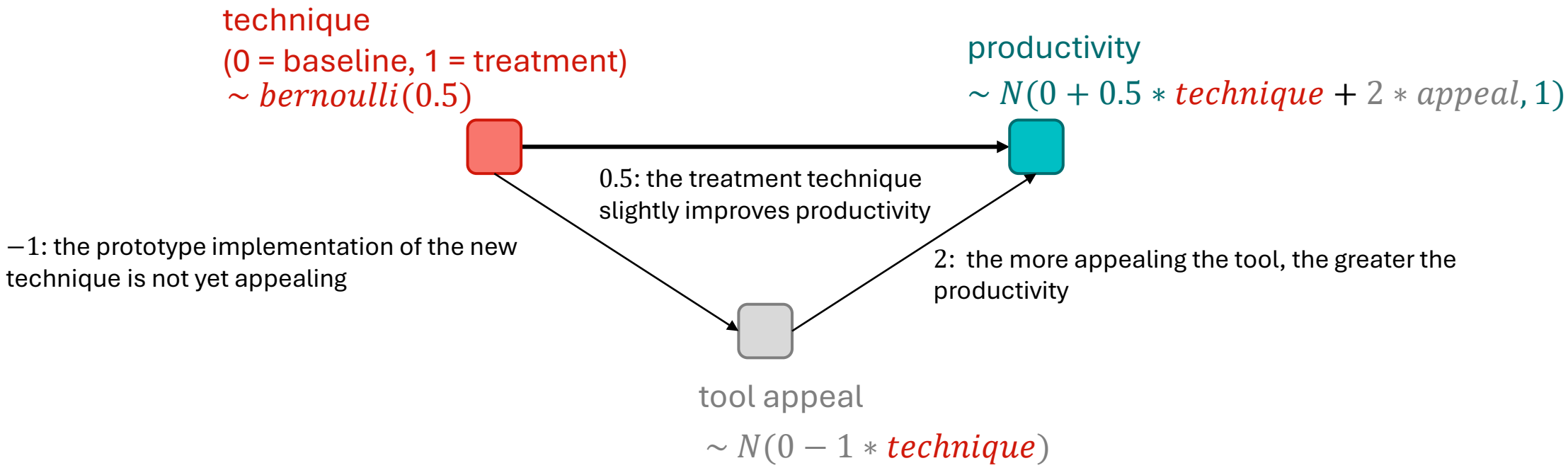
Inozemtseva, L., & Holmes, R. (2014, May). Coverage is not strongly correlated with test suite effectiveness. In *Proceedings of the 36th international conference on software engineering* (pp. 435-445).

Most Influential Paper at ICSE’24

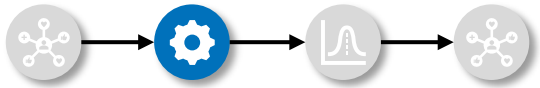
Mediation: Pipes



"The company tested a new **technique** in their requirements engineering phase that is enabled by a **prototype tool**. We investigated whether this technique improved the engineers' **productivity**."



Mediation: Pipes



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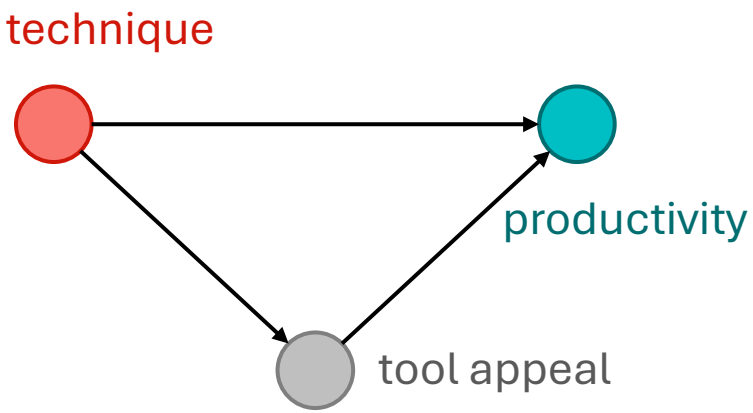
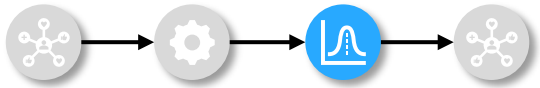


Model Derivation

m_1: $productivity \sim 1 + technique$

m_2: $productivity \sim 1 + technique + appeal$

Mediation: Pipes



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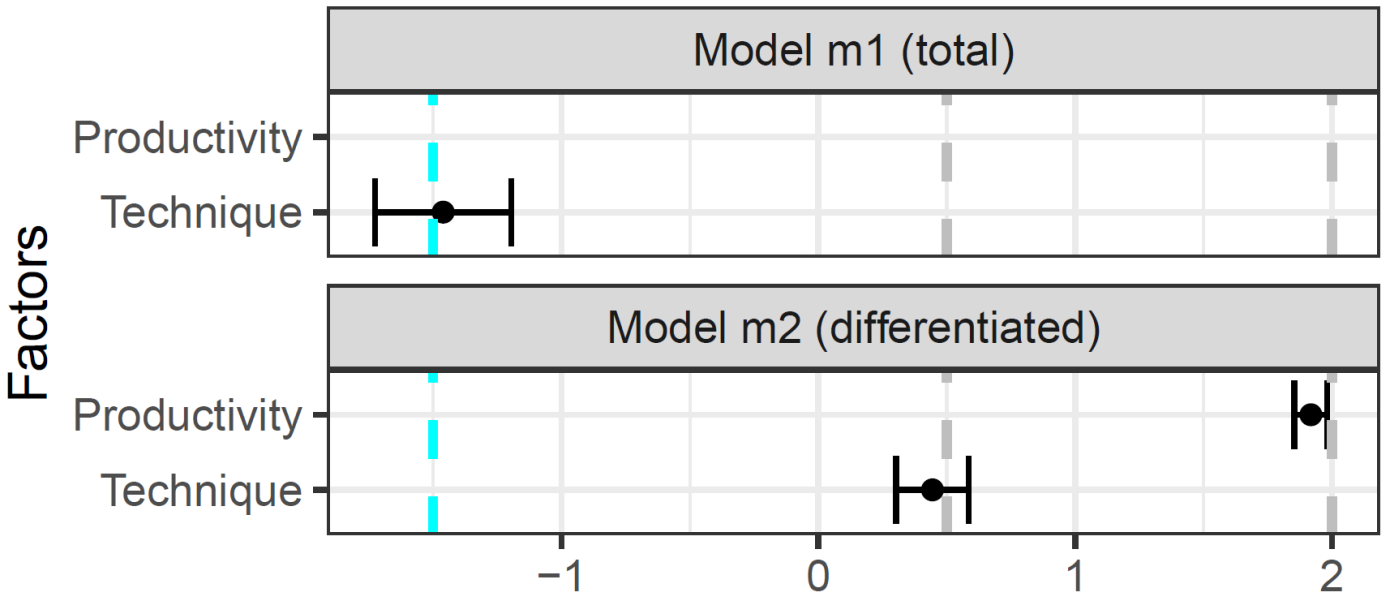
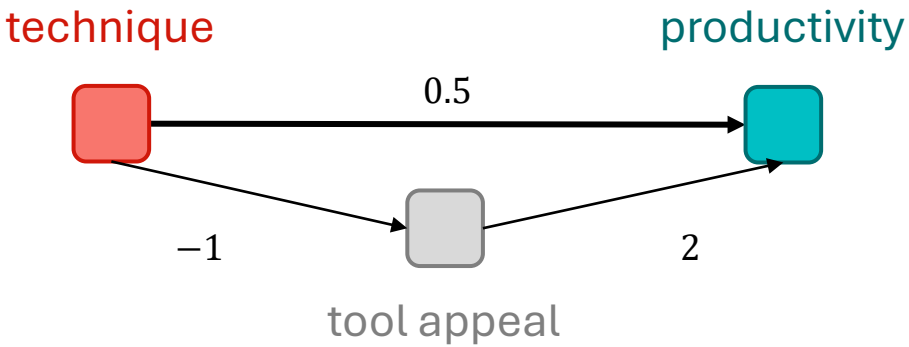


$\sim N(0 - 1 * technique)$



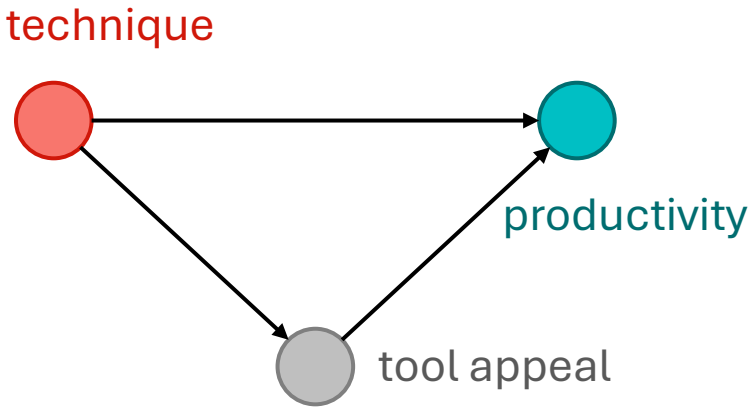
Direct effect: 0.5

Indirect effect: $-1 * 2 = -2$

Total effect: $0.5 - 2 = -1.5$

Mediation: Pipes

Both statistical models are causally unbiased. The choice, which of them to prefer depends on whether one is interested in the direct or total effect of the treatment.



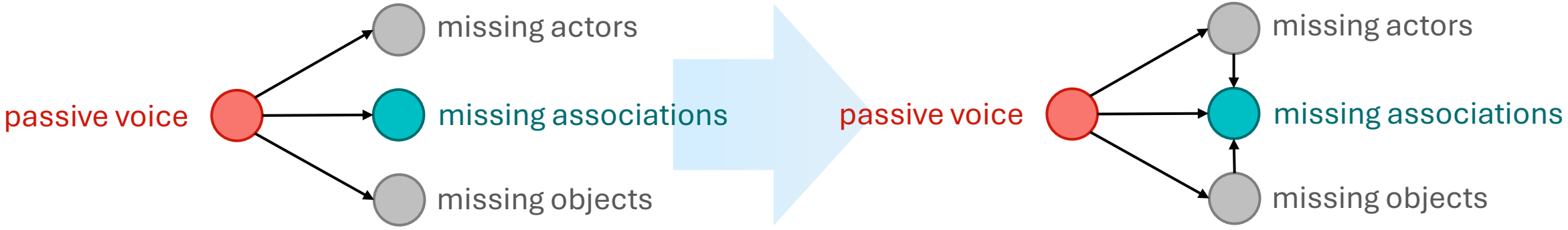
Model Derivation

m_1: $productivity \sim 1 + technique$

m_2: $productivity \sim 1 + technique + appeal$

> If the effect of the **treatment** on the **mediator** is unstable, it is advisable to differentiate the direct from the indirect effect (i.e., $m_2 > m_1$).

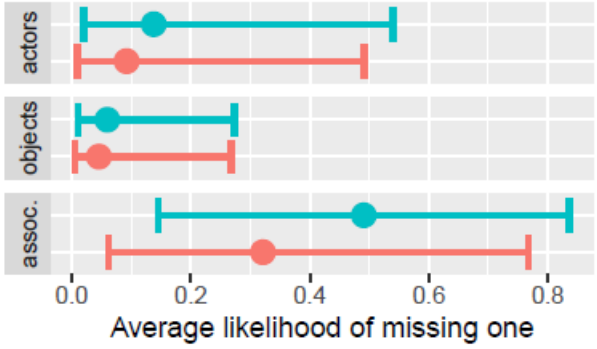
Practical Example of Mediation



Mann-Whitney test:

passive voice → missing associations

$$p = 0.03 < 0.05 = \alpha$$



https://doi.org/10.1145/2652524.2652554

Femmer, H., Kučera, J., & Vetrò, A. (2014, September). On the impact of passive voice requirements on domain modelling. In *Proceedings of the 8th ACM/IEEE international symposium on empirical software engineering and measurement* (pp. 1-4).

https://doi.org/10.1145/3643664.3648211

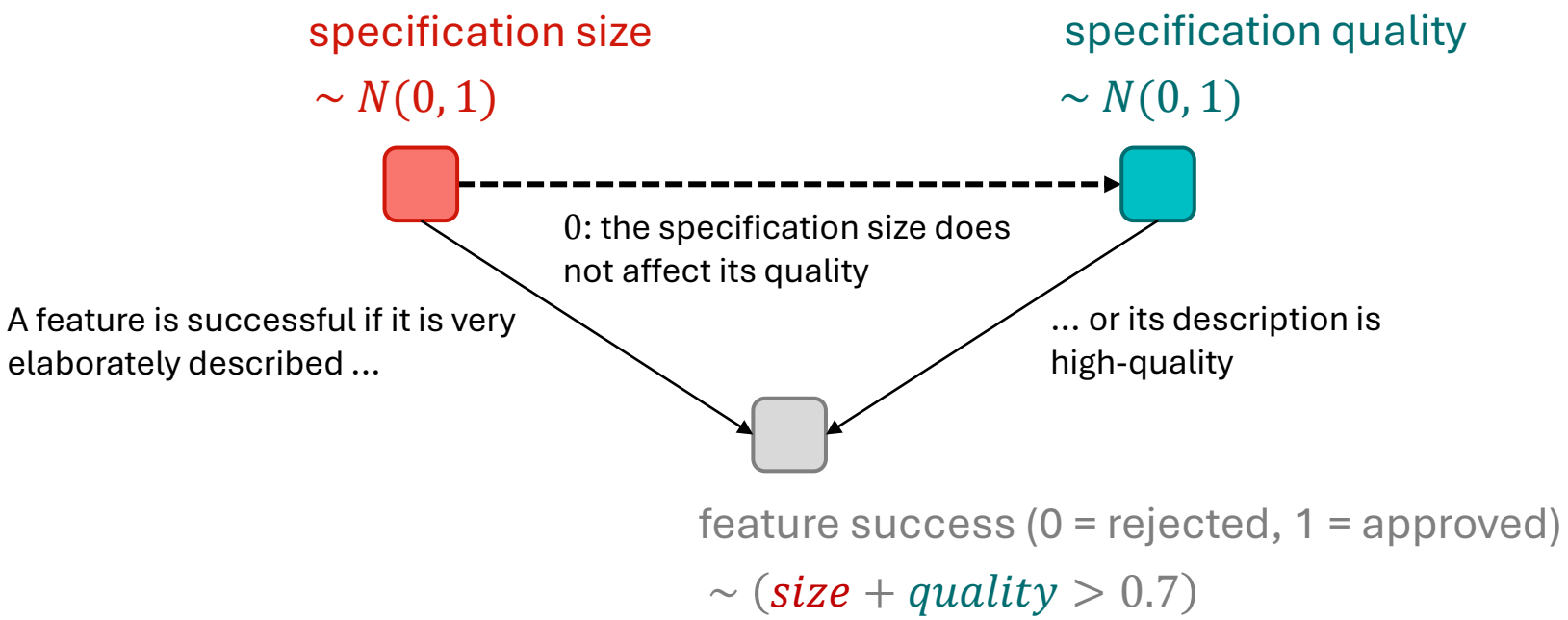
Frattini, J., Fucci, D., Torkar, R., & Mendez, D. (2024, April). A second look at the impact of passive voice requirements on domain modeling: Bayesian reanalysis of an experiment. In *Proceedings of the 1st IEEE/ACM International Workshop on Methodological Issues with Empirical Studies in Software Engineering* (pp. 27-33).

Causal Inference II

Colliders, D-separation, and Model Comparison

Colliding: Common Effects

"Does the length of a requirements specification affect its quality? Both factors do affect whether the specified feature is implemented successfully."



Colliding: Common Effects

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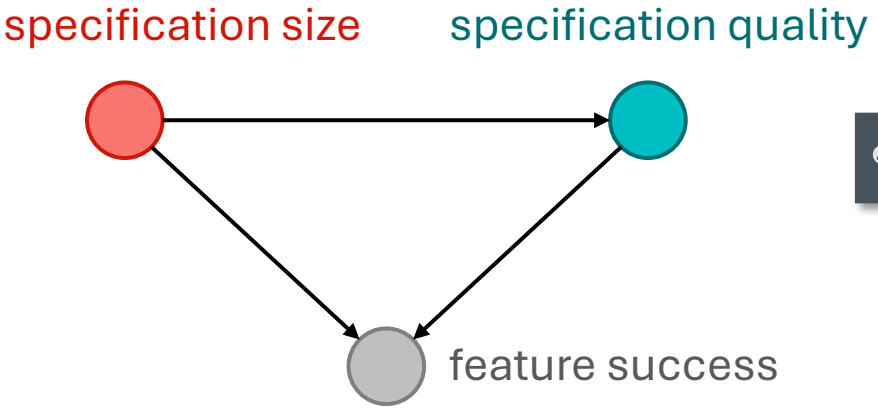
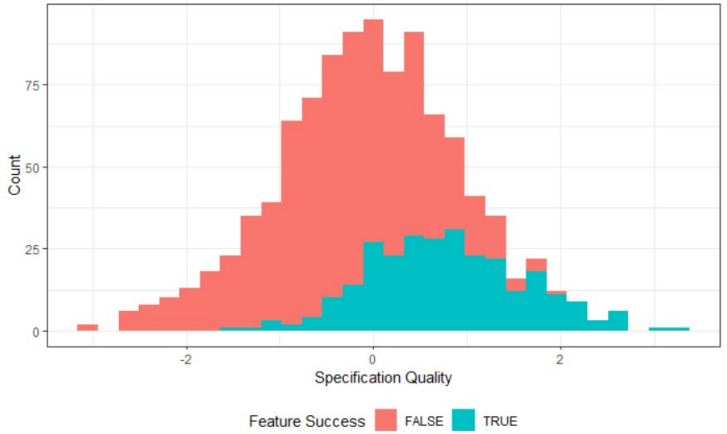


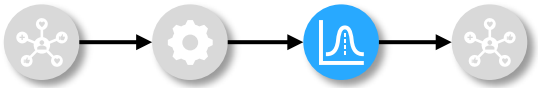
Model Derivation

m_1: $quality \sim 1 + size + success$

m_2: $quality \sim 1 + size$



Colliding: Common Effects



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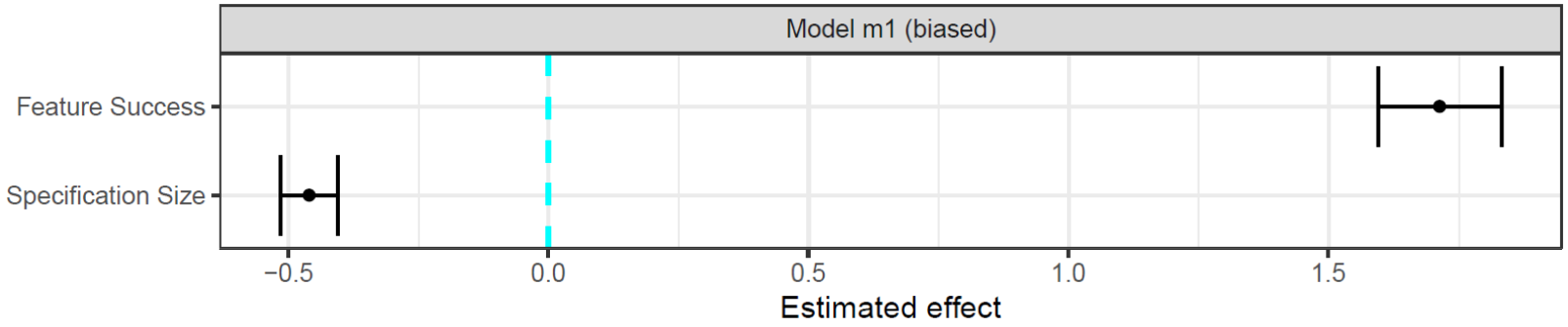
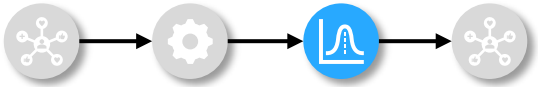


Colliding: Common Effects



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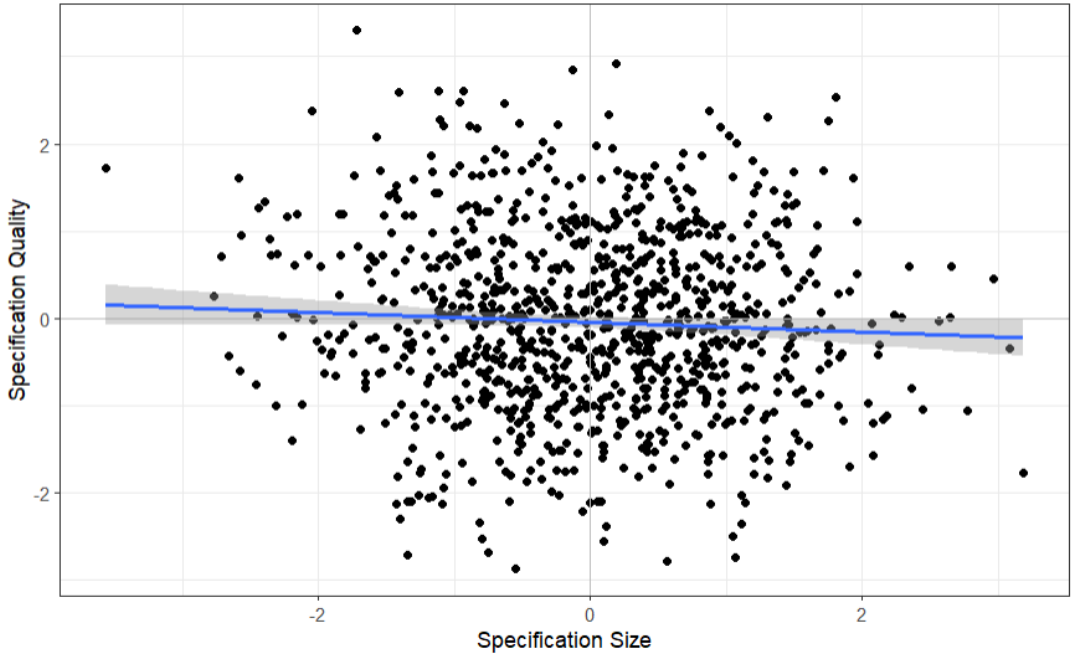
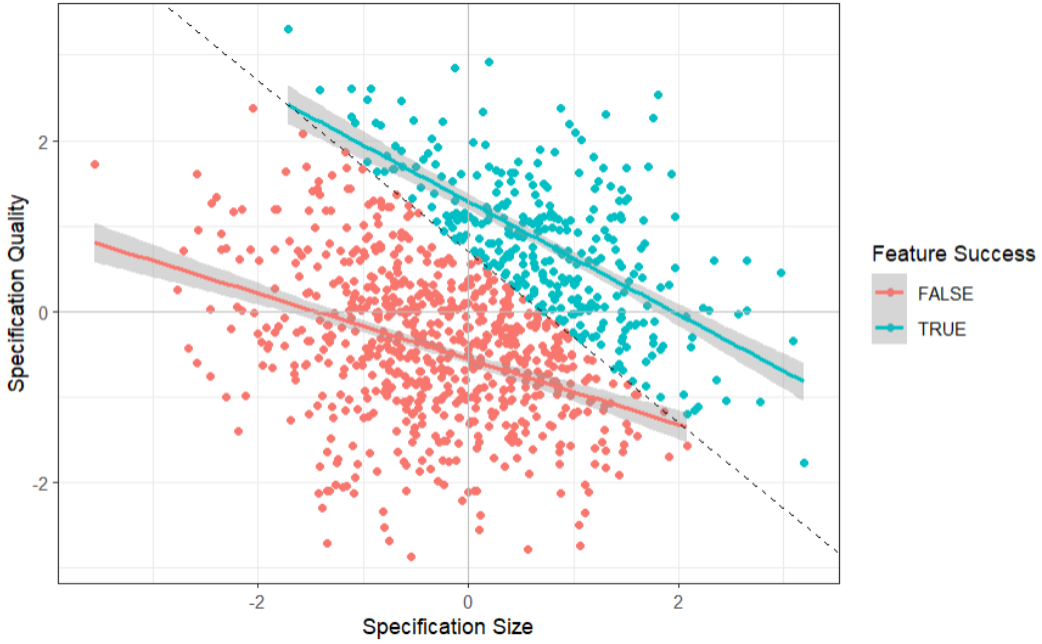
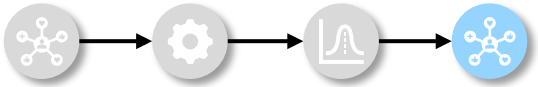




Colliding: Common Effects



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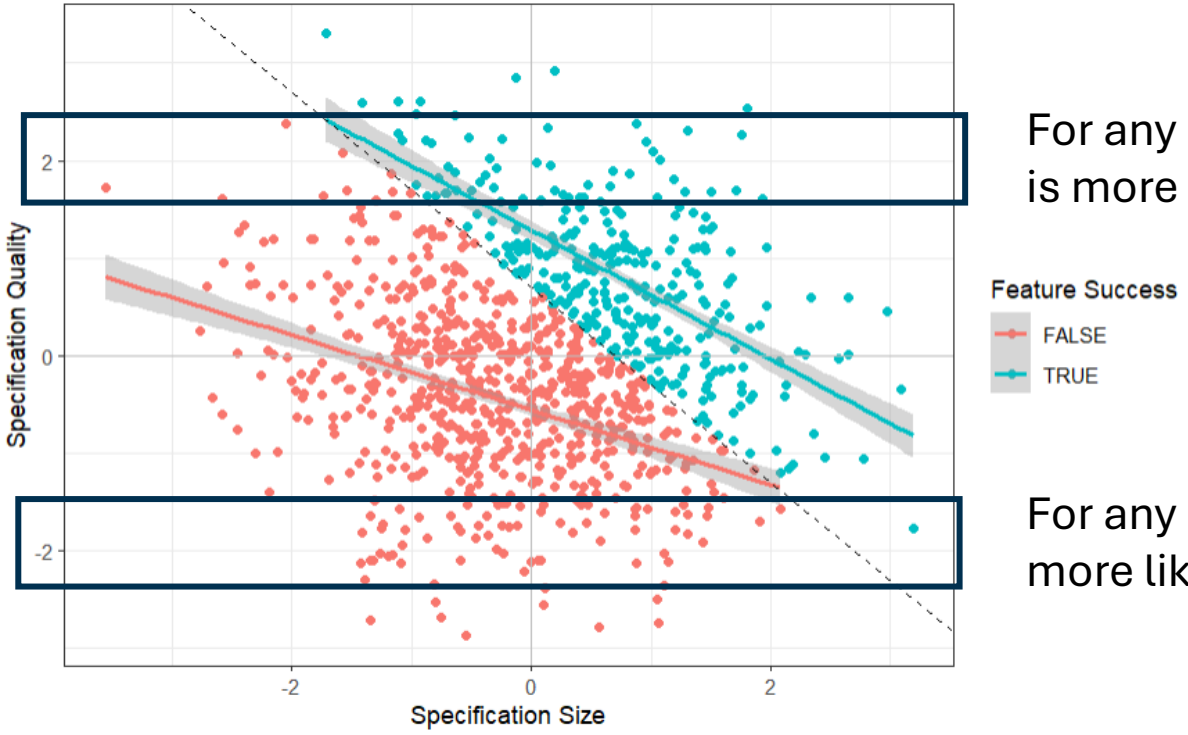
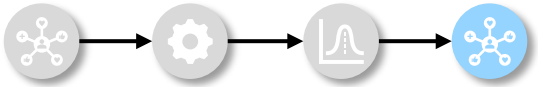


For any high quality value, if feature success is *false*, then it is more likely that the specification size was small.

For any low quality value, if feature success is *true*, then it is more likely that the specification size was large.

Colliding: Common Effects



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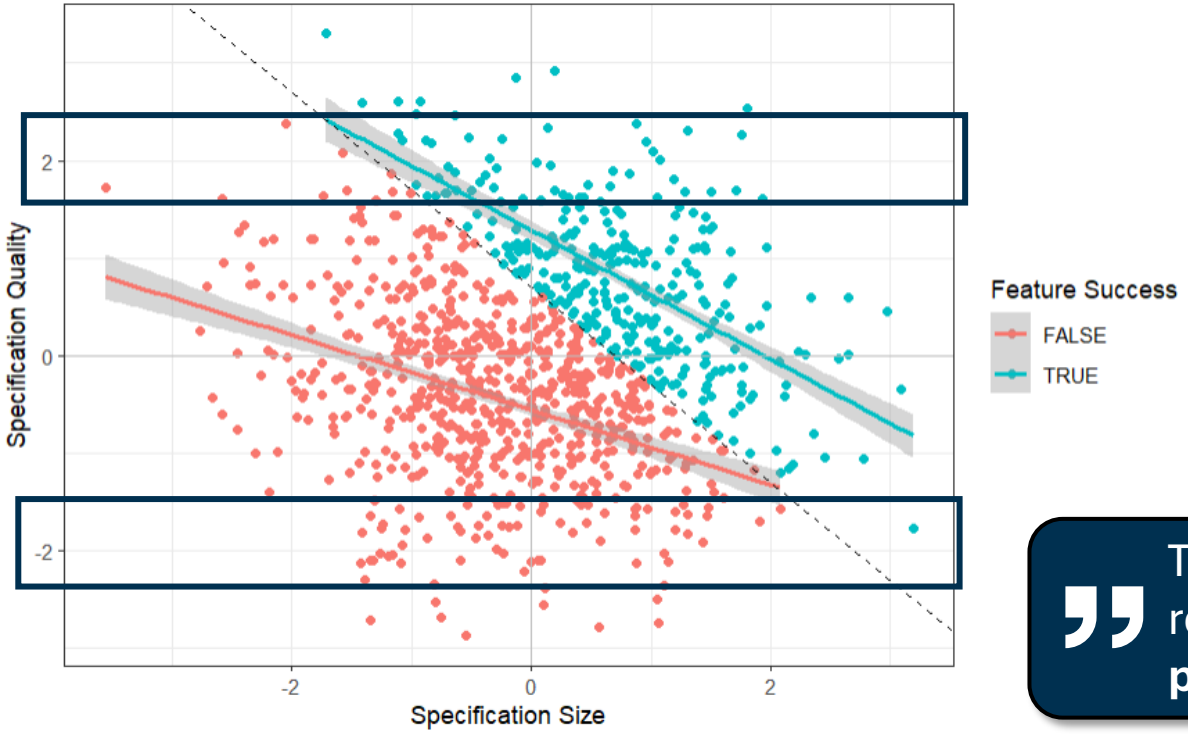
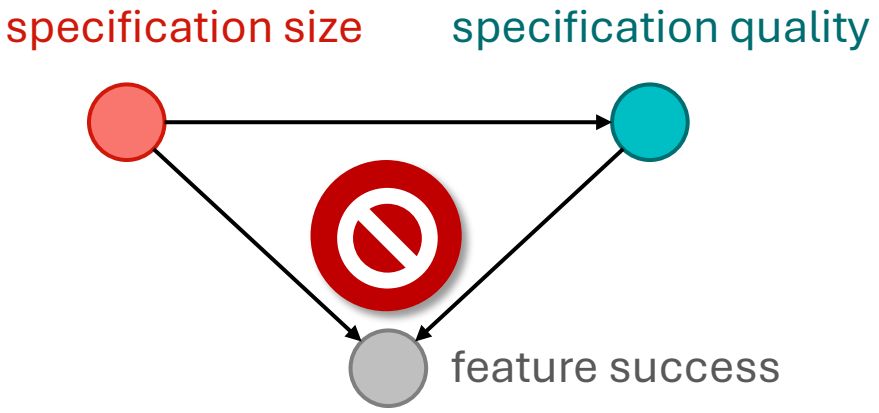
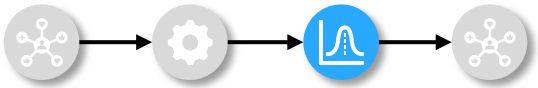




> The non-causal association suggests a negative relationship between the cause and effect (**Berkson's paradox/fallacy**).

Colliding: Common Effects



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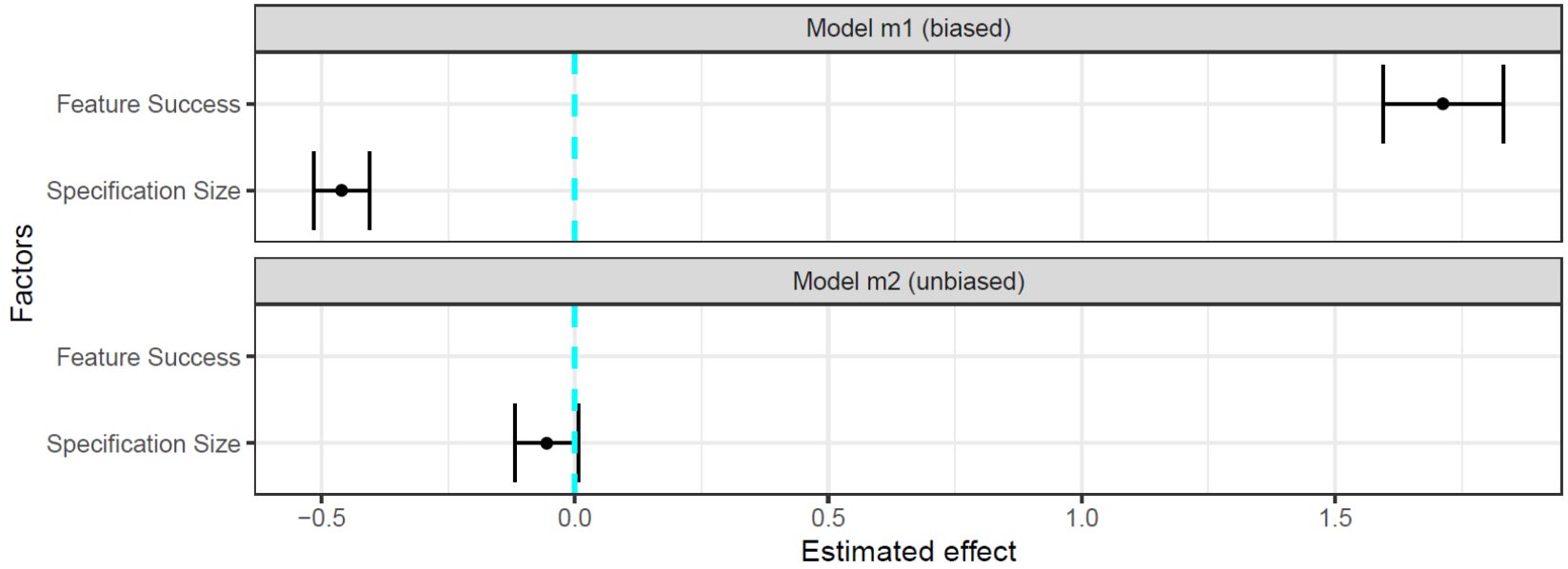

$m_1: quality \sim 1 + size + success$

$m_2: quality \sim 1 + size$

Colliding: Common Effects

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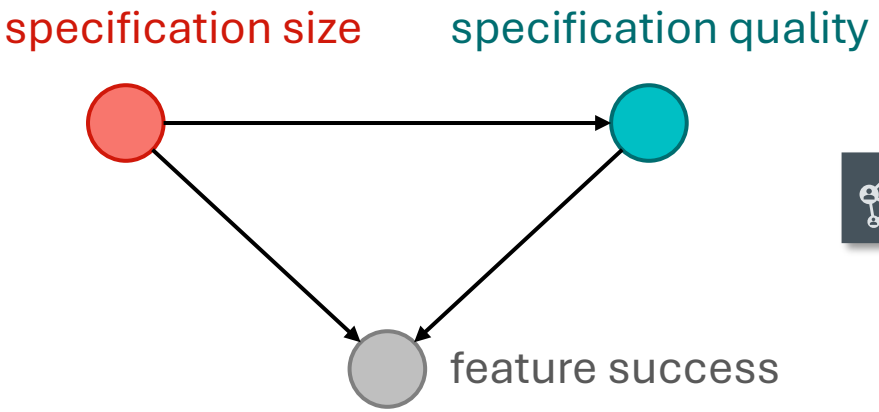
"Does the length of a requirements specification affect its quality? Both factors do affect whether the specified feature is implemented successfully."



Model Derivation

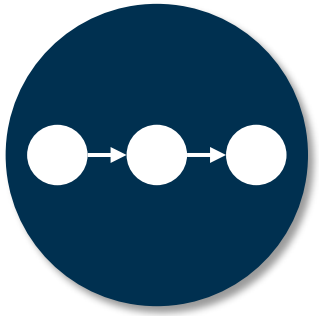
$\cancel{m_1: quality \sim 1 + size + success}$

$m_2: quality \sim 1 + size$

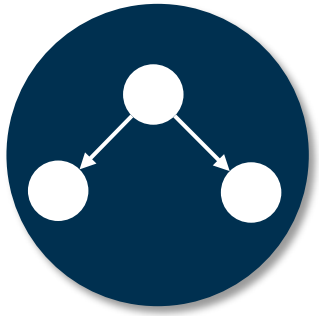
> If both treatment and outcome have a common effect, we must not condition on this common effect. Consequently, we cannot simply add *all* relevant variables as predictors.

Basic Types of Association

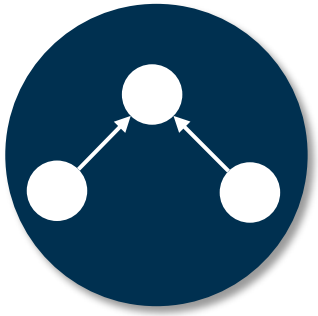
There are three basic types of association



Mediator



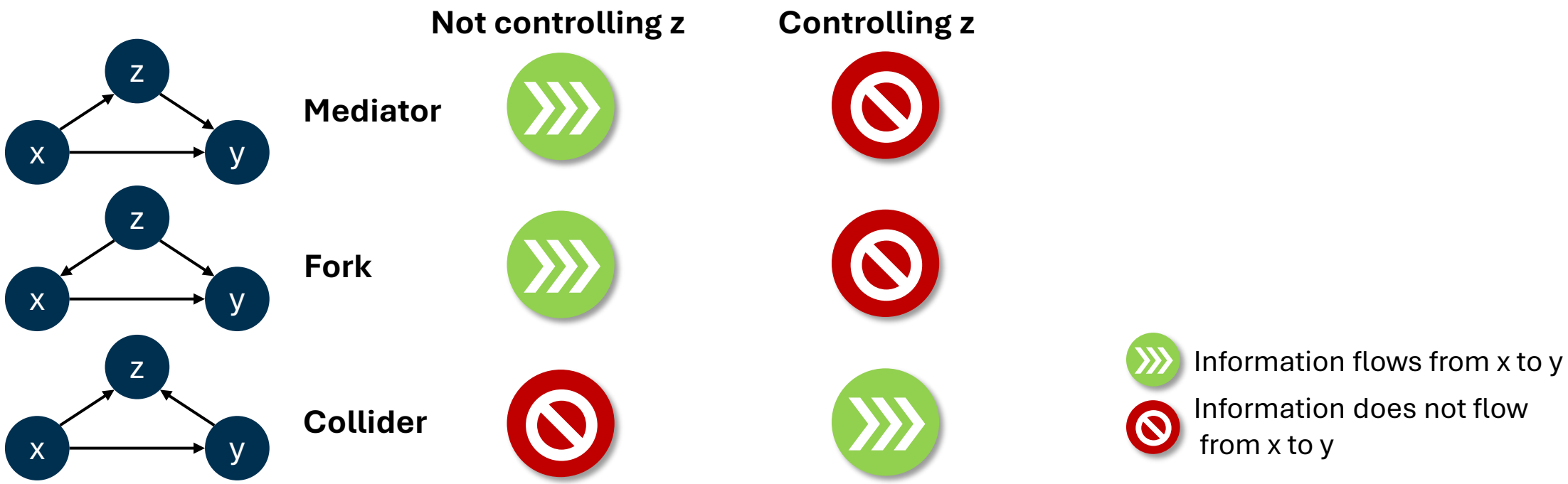
Fork



Collider

Basic Types of Association

The three basic types of association behave differently when statistically controlling for the third variable:



Causal Modelling

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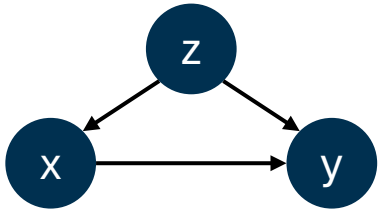
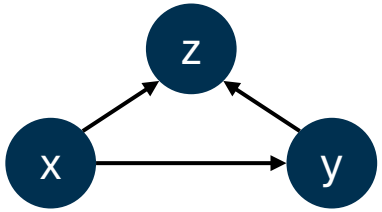
Causal modelling is imperative for drawing the correct conclusions when conducting statistical causal inference.

> We cannot infer causality from data alone – we must also know the process that created that data.

Causal Model



Fork



Collider

Model Derivation

x	y	z

Statistical Model

$m: y \sim x + z$

$m: y \sim x$

? But how does this approach scale beyond three variables?

Deriving a statistical Model from a causal Model: d-separation

A general framework to derive statistical models from causal models uses **d-separation**. It produces an **adjustment set**, i.e., the set of variables to control statistically in order to de-confound the causal effect between the exposure and the outcome.

https://bayes.cs.ucla.edu/BOOK-2K/d-sep.html

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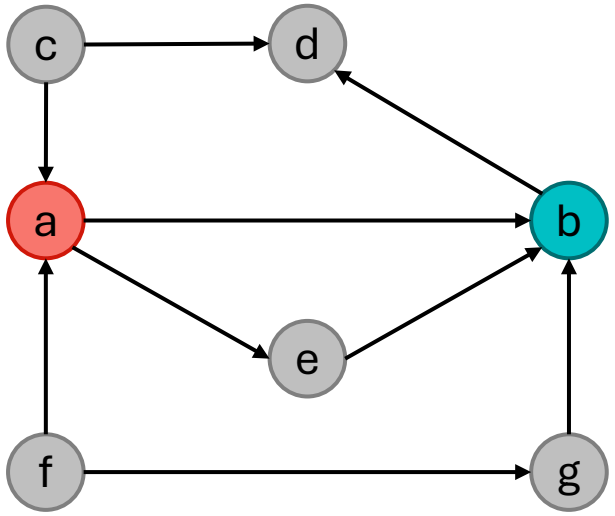

Model Derivation

$m: b \sim a$

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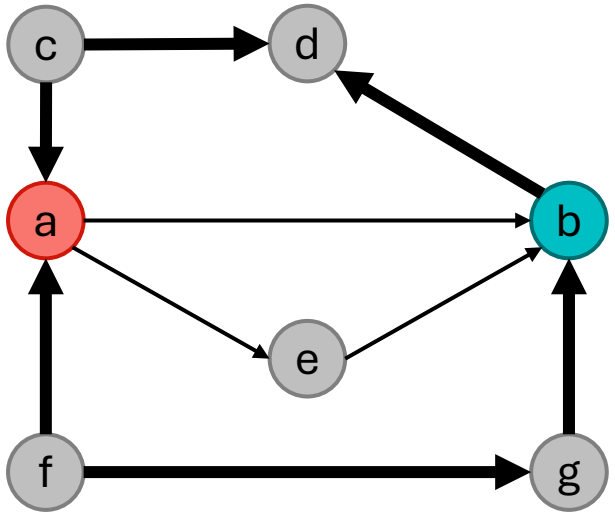

Model Derivation

$m: b \sim a$

1. Identify every **backdoor path**.

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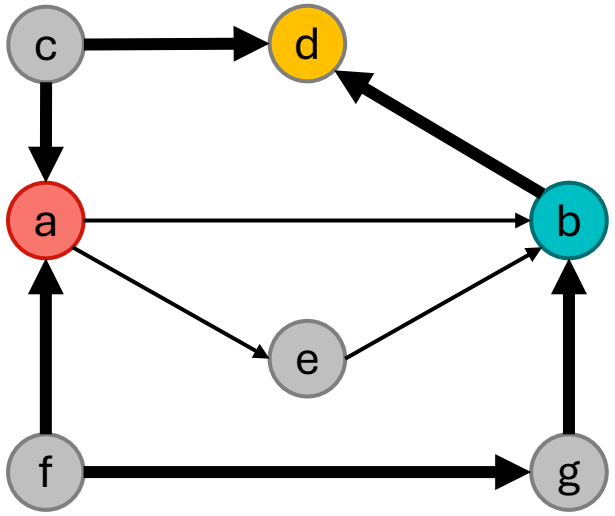

Model Derivation

$m: b \sim a$

1. Identify every backdoor path.
2. Check if there is a **collider** between the cause and effect of interest.
   1. If yes, do not condition on any of the variables on this path.

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Deriving a statistical Model from a causal Model: d-separation

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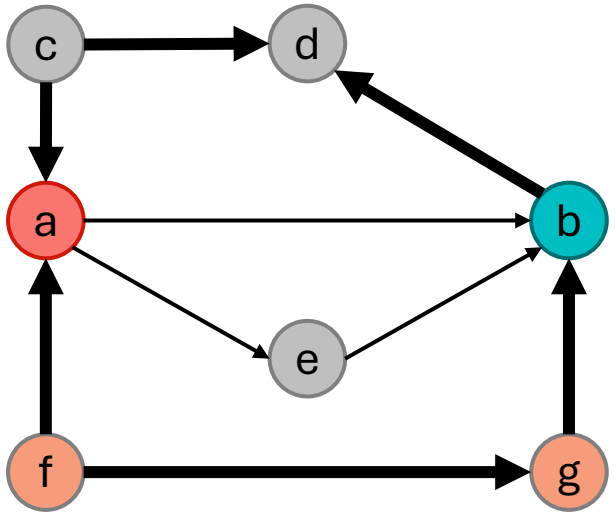


Model Derivation

$m: b \sim a + f$

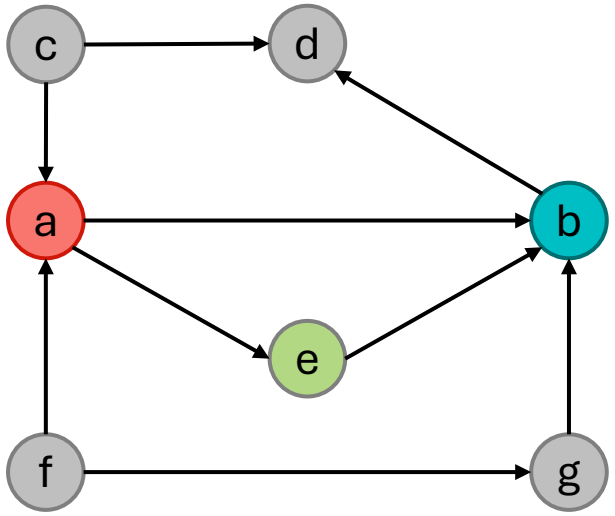
$m: b \sim a + g$

1. Identify every backdoor path.
2. Check if there is a collider between the cause and effect of interest.
   1. If yes, do not condition on any of the variables on this path.
   2. If not, condition on **any of the variables** on this path to "close" the backdoor path.

https://bayes.cs.ucla.edu/BOOK-2K/d-sep.html

Deriving a statistical Model from a causal Model: d-separation

A general framework to derive statistical models from causal models uses **d-separation**. It produces an **adjustment set**, i.e., the set of variables to control statistically in order to de-confound the causal effect between the exposure and the outcome.



Model Derivation

$m: b \sim a + f(+e)$

$m: b \sim a + g(+e)$

1. Identify every backdoor path.
2. Check if there is a collider between the cause and effect of interest.
   1. If yes, do not condition on any of the variables on this path.
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3. Optionally, condition on **mediators** to discern the direct effect.

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Deriving a statistical Model from a causal Model: d-separation

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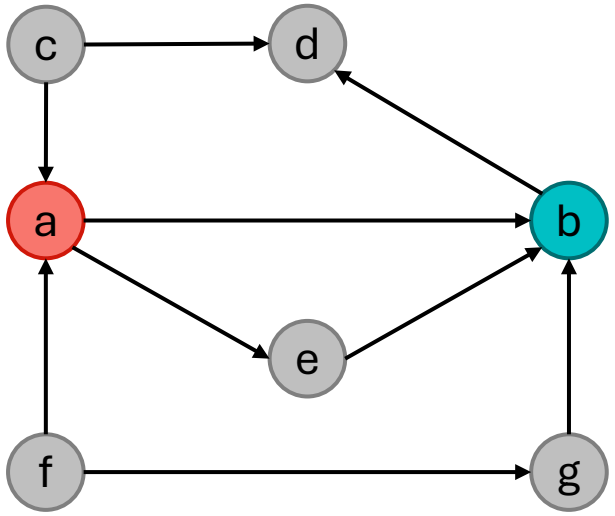

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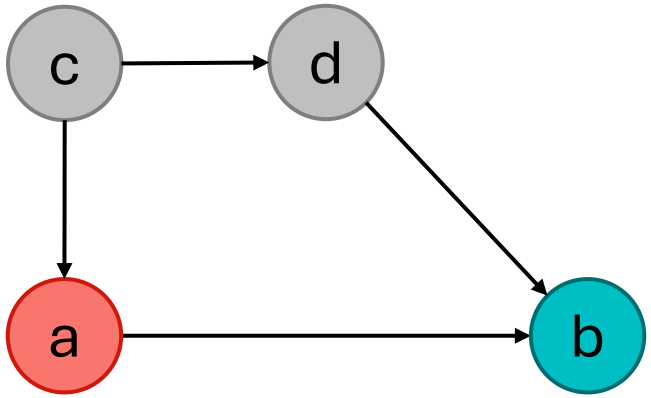
   2. If not, condition on any of the variables on this path to "close" the backdoor path.
3. Optionally, condition on mediators to discern the direct effect.

https://bayes.cs.ucla.edu/BOOK-2K/d-sep.html

Exercises

Determine the **adjustment set** of variables to de-confound the estimation of the effect of the phenomenon a → b via d-separation.



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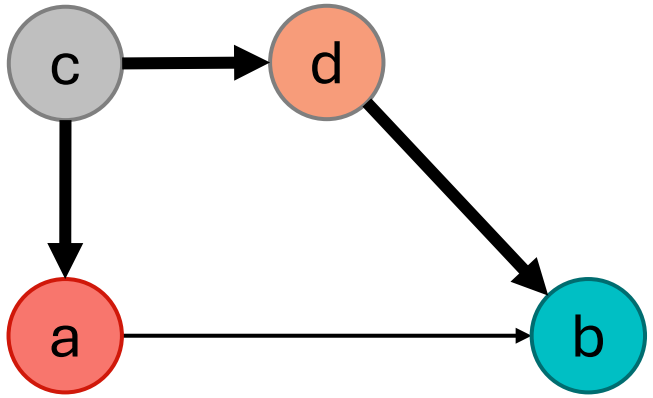
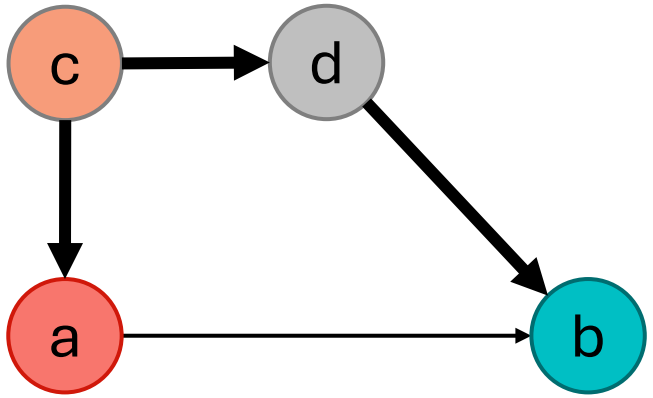


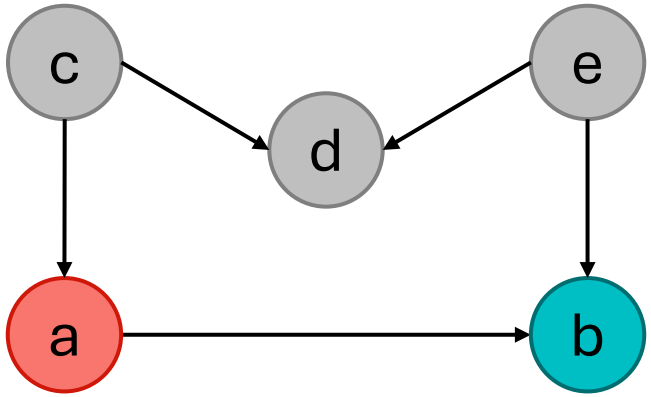


Adjustment set = {d}

Adjustment set = {c}

Exercises

Determine the **adjustment set** of variables to de-confound the estimation of the effect of the phenomenon a → b via d-separation.



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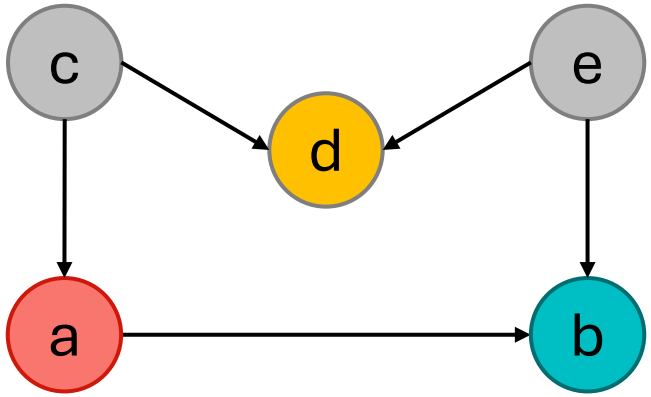

Adjustment set = {}

Exercises

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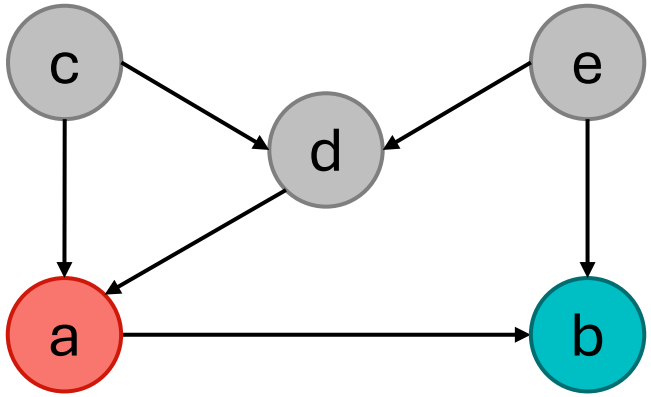

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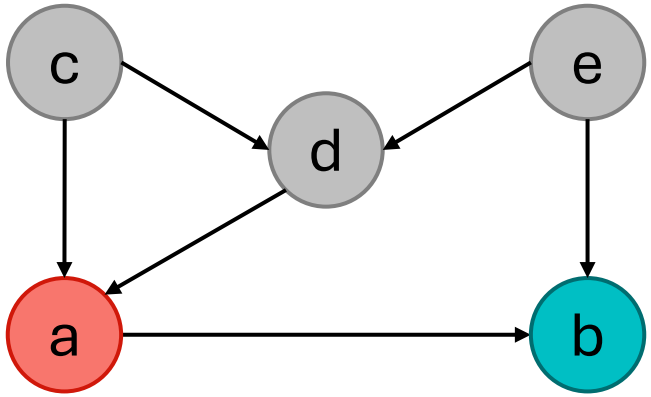
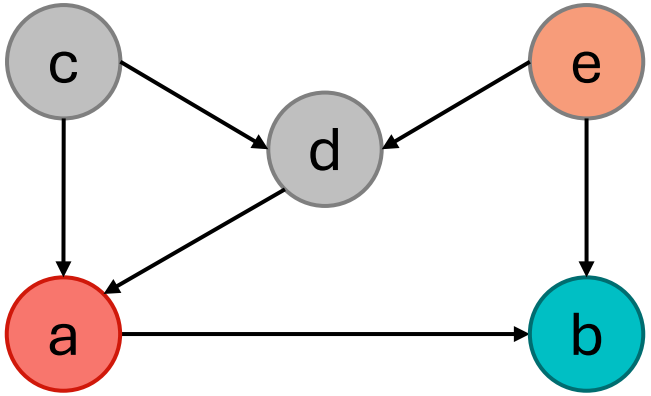


The DAG contains two backdoor paths:

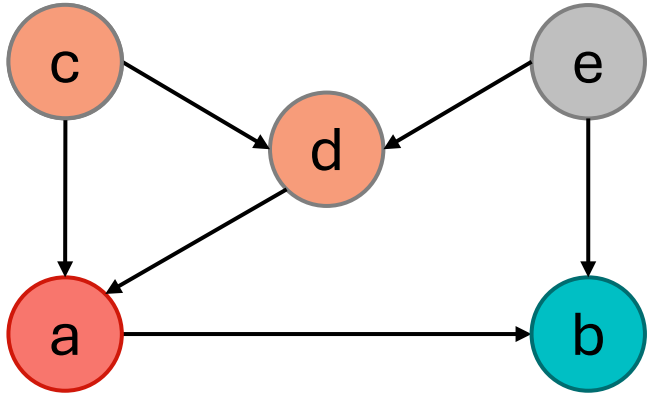
1. a ← c → d ← e → b
2. a ← d ← e → b

Exercises

Determine the **adjustment set** of variables to de-confound the estimation of the effect of the phenomenon a → b via d-separation.



Adjustment set = {e}

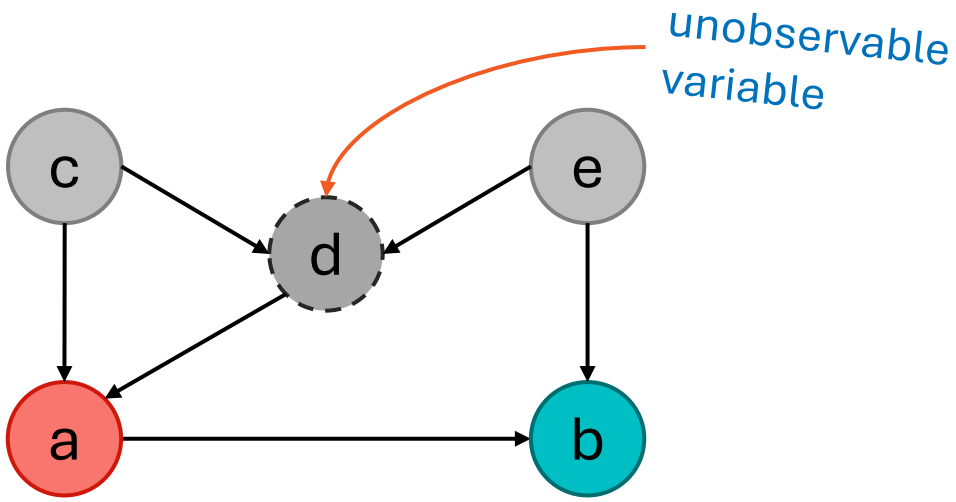


Adjustment set = {c, d}

> Conditioning on d opens the backdoor-path a ← c → d ← e → b which then has to be closed by additionally conditioning on c.

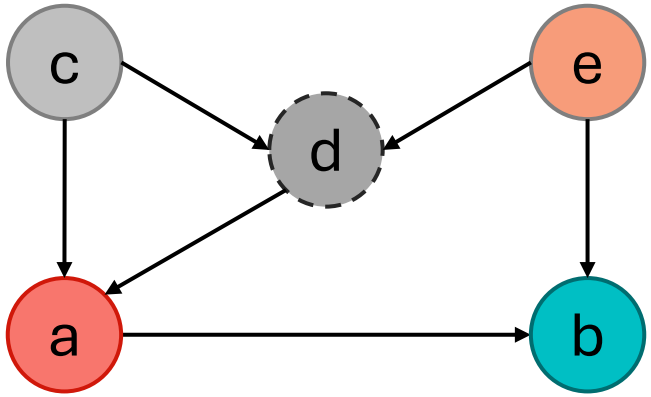
Exercises

Determine the **adjustment set** of variables to de-confound the estimation of the effect of the phenomenon a → b via d-separation.



Exercises

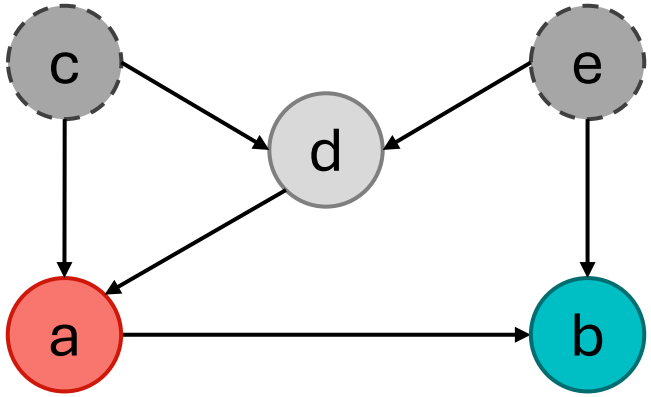
Determine the **adjustment set** of variables to de-confound the estimation of the effect of the phenomenon a → b via d-separation.



Adjustment set = {e}

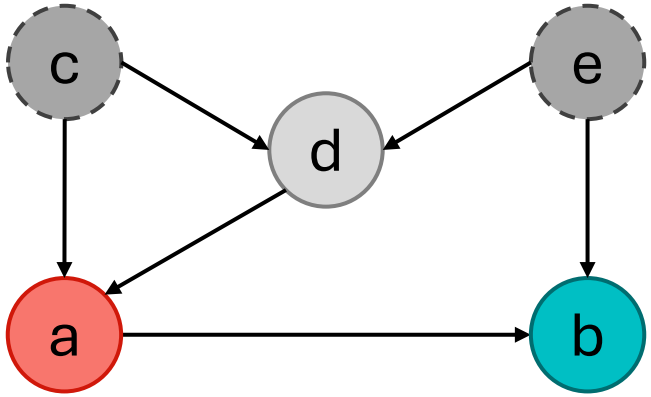
Exercises

Determine the **adjustment set** of variables to de-confound the estimation of the effect of the phenomenon a → b via d-separation.



Exercises

Determine the **adjustment set** of variables to de-confound the estimation of the effect of the phenomenon a → b via d-separation.



> For some DAGs, it is impossible to find an adjustment set that deconfounds the effect of interest.

Causal Modelling

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d-separation is one reliable technique for Model Derivation but still depends on the construction of a “correct” causal model, i.e., a DAG representing causal assumptions where

- the included variables are **complete**, and
- the included (and excluded) relationships between them are **correct**.

? But in reality there is no grand reveal of the “actual” causal model (as in the simulations). So how do you determine which model is “correct”?

Causal Modelling

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It is impossible to determine the *correct* causal model – we would need to look behind the curtain and see the ”fabric of the world.” But instead of aiming for a *correct* causal model, rather aim for a **transparent** and **useful** causal model.

Transparent
Visualize causal assumptions via DAGs to allow readers the underlying causal model

Useful
?

Model Comparison

> Depending on the observed change in results, you can infer which assumed causal model describes the data creation process best

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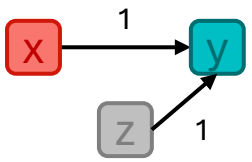
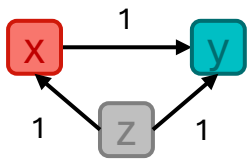
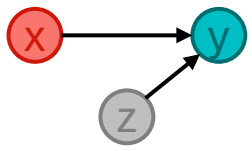
Simulated ground-truth

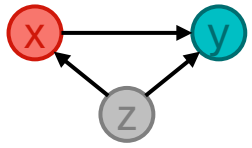




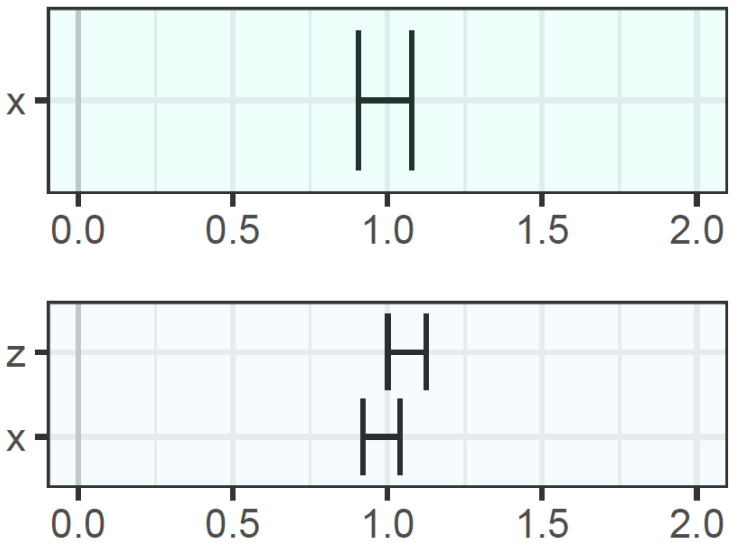
Assumed causal (and consequent statistical) model

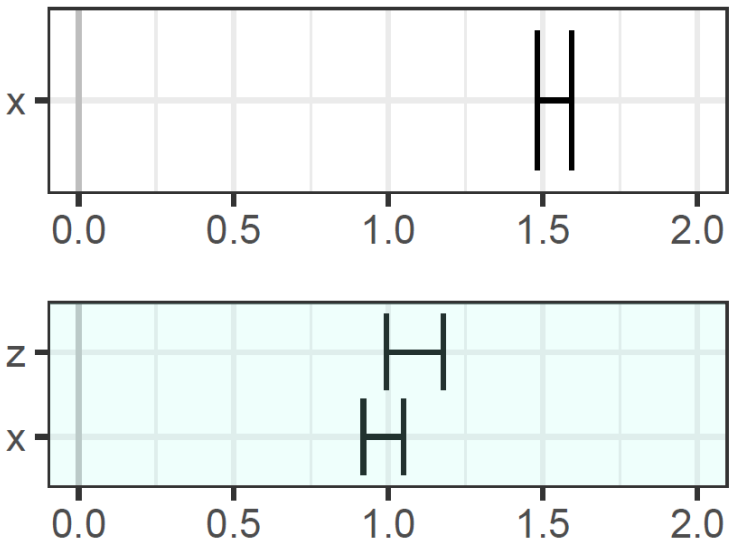


$y \sim x$



$y \sim x + z$





If the effect of x on y does not when including z, then z has **no** causal effect on x.

If the effect of x on y changes when including z, then z has a causal effect on x.

Model Comparison

We cannot assess the usefulness of a model in absolute terms, but in relative terms via **model comparison**!

Useful
Reflecting the data generation process more accurately

> A model that reflects the data generation process more accurately will provide more robust inferences under differing conditions. It is therefore – at least relatively speaking – more useful.

Statistical Causal Inference for Requirements Engineering

An Application Example of SCI in RE

Heyn, H. M., Mao, Y., Weiss, R., & Knauss, E. (2025, June). Causal Models in Requirement Specifications for Machine Learning: A vision. In *Proceedings of the 33rd ACM International Conference on the Foundations of Software Engineering* (pp. 1402-1405).

Causal Models in Requirements Specifications for Machine Learning

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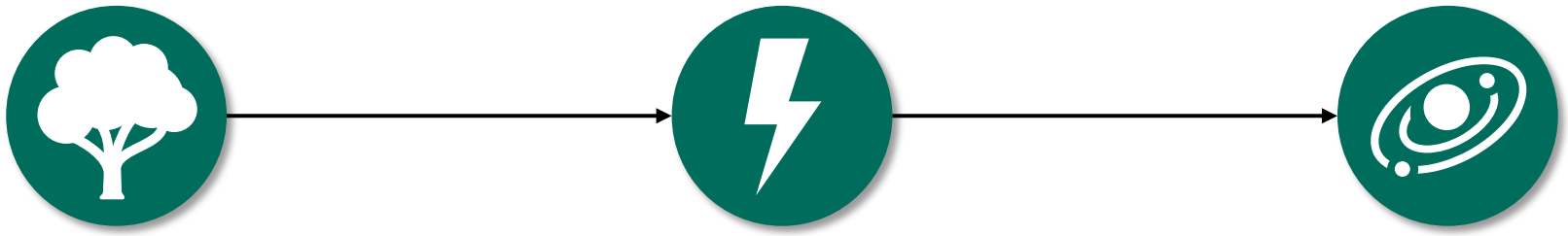
RE struggles to translate high-level requirements into data needs. One reason for this under-specification is that operational context is often **implicitly assumed in the data collection**.



Operational Context
implicitly assumed in data collection

Brittle Models
that fail to generalize

Low Transferability
into unknown contexts

Ahmad, K., Abdelrazek, M., Arora, C., Bano, M., & Grundy, J. (2023). Requirements engineering for artificial intelligence systems: A systematic mapping study. *Information and Software Technology*, *158*, 107176.
Mitchell, S., Potash, E., Barocas, S., D'Amour, A., & Lum, K. (2021). Algorithmic fairness: Choices, assumptions, and definitions. *Annual review of statistics and its application*, *8*(1), 141-163.
D'Amour, A., Heller, K., Moldovan, D., Adlam, B., Alipanahi, B., Beutel, A., ... & Sculley, D. (2022). Underspecification presents challenges for credibility in modern machine learning. *Journal of Machine Learning Research*, *23*(226), 1-61.

Causal Modelling in RE

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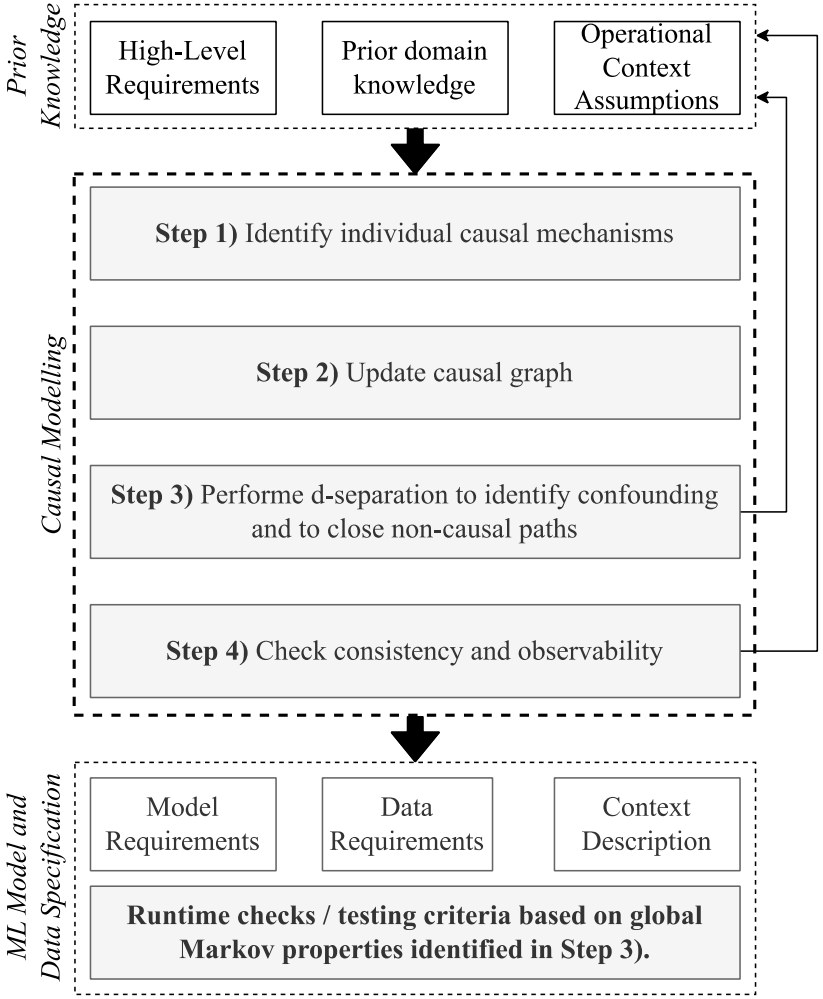
Using the principles of SCI in RE allows to …

- derive data and model requirements from prior knowledge about the operational context.
- enable documentation of these prior assumptions.
- allow for a systematic determination of testable conditions.
- **use of causal modelling as a Requirement Engineering activity.**

Causal RE workflow



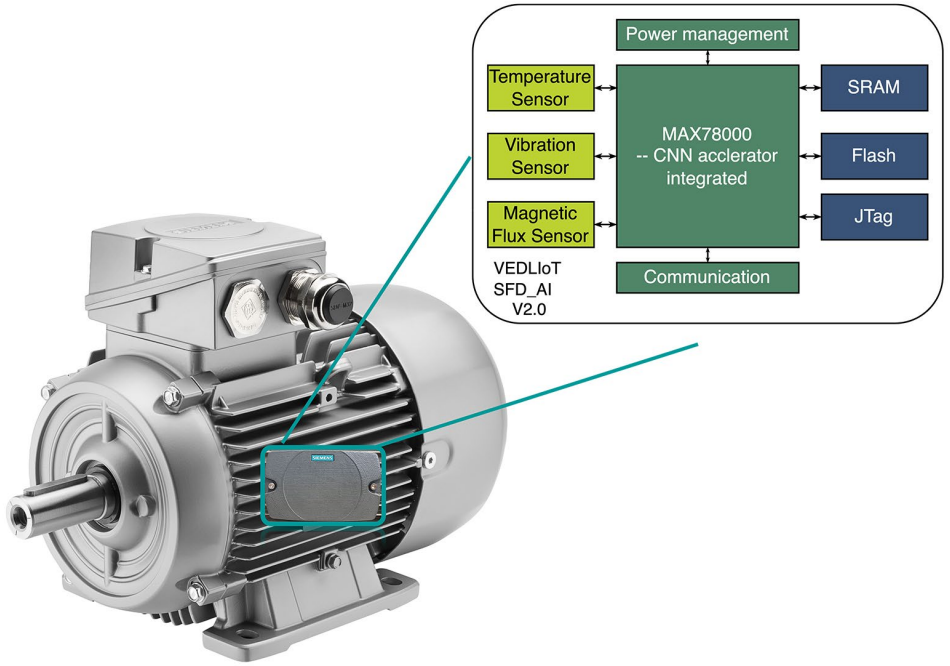
- **Step 1:** Identify individual causal mechanisms
- **Step 2:** Create / update a graphical causal model (DAG)
- **Step 3:** Use d-separation to identify confounding and close non-causal paths.
- **Step 4:** Check graph consistency and observability of necessary variables.

Use case example

Context: ML-based classification (outcome) of cooling faults in electric motors (exposure). The system uses sensors for temperature, vibration, and magnetic flux.

Functional Requirement:

GIVEN indoor operational environment
WHEN the cooling system is abnormal
THEN an alarm should be raised.



Methodology

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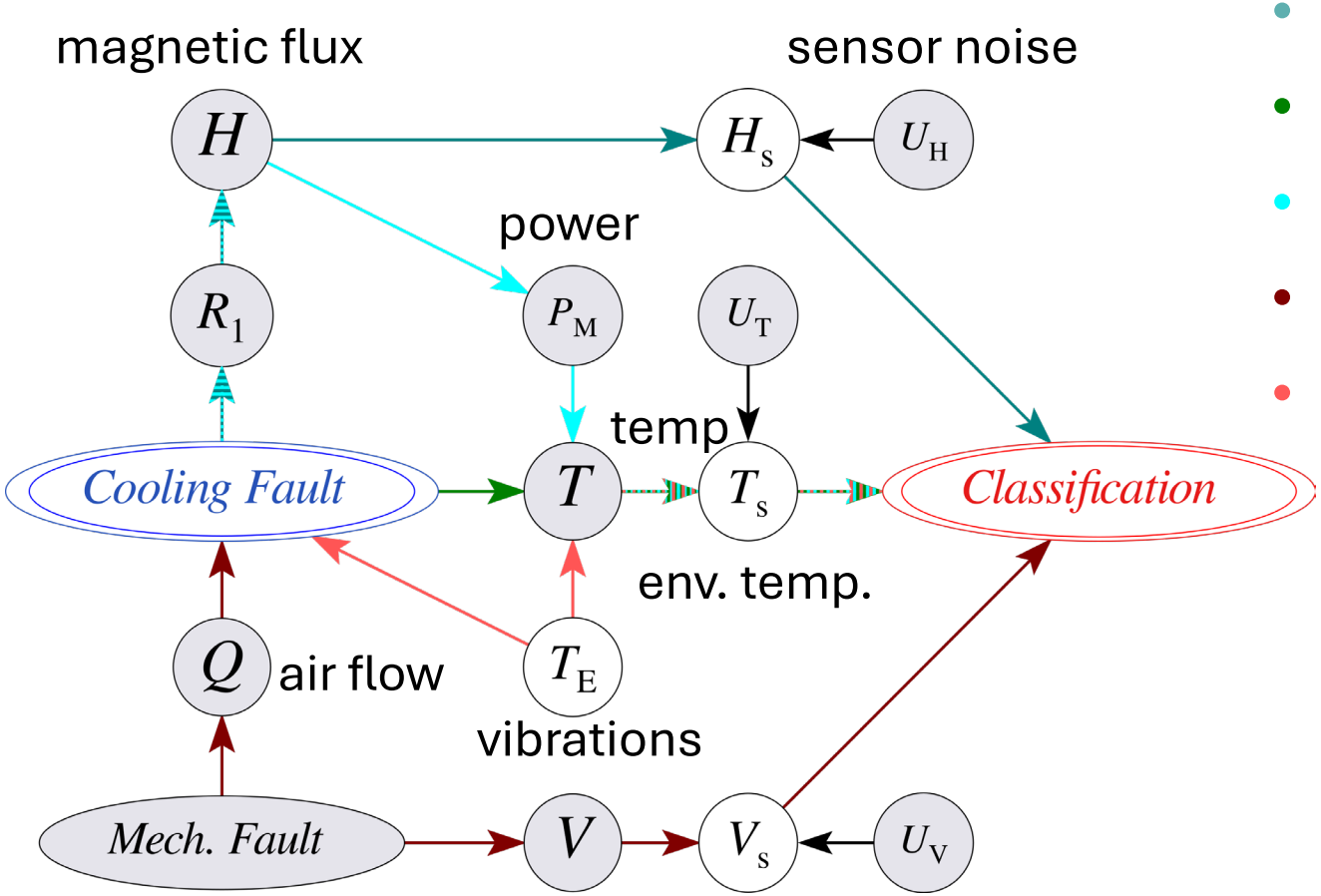
- Three exploratory workshops with two Siemens engineers and two academic researchers.
- We introduced to the concept of causal modelling and d-separation.
- Company expert outlined the use case and prior knowledge was formalized by identifying causal mechanisms and updating the causal model.

Interesting observation:

The structured approach to formalizing prior knowledge as causal graphs allowed them to ”discover” more prior knowledge.

- We used d-separation to close non-causal paths between exposure (i.e., the occurrence of a cooling fault), and the outcome (i.e., the classification result).
- This resulted in data and model requirements, as well as testable conditions.

Resulting graphical causal model

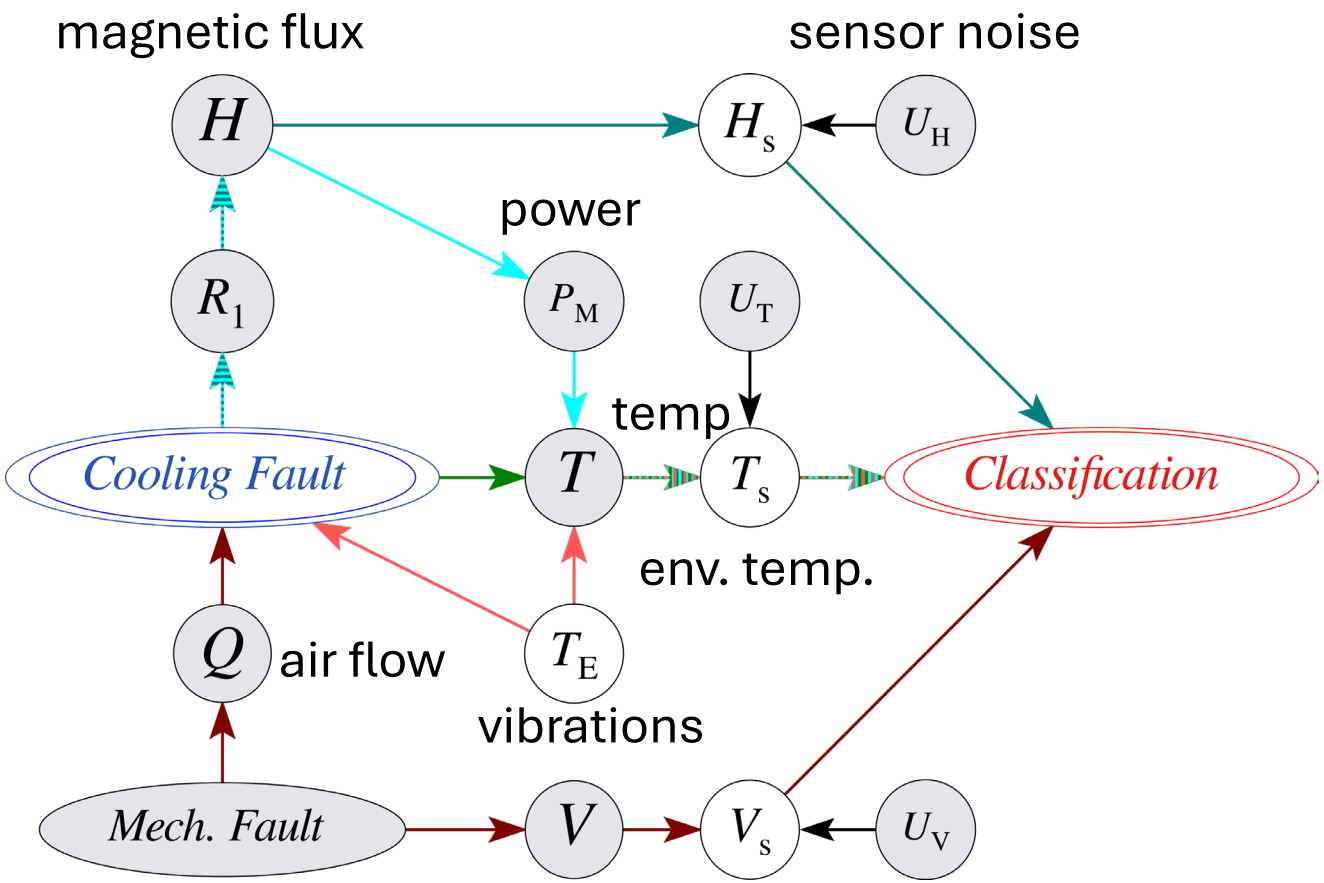


- Magnetic flux mechanism
- Temperature mechanism
- Mechanical power mechanism
- Mechanical fault confounding
- Environmental temperature confounding

Observations:

- Vibration data alone is insufficient to detect cooling faults, because not all mechanical faults lead to a cooling fault.
- Data on temperature and magnetic flux should be included.
- Sensor data should contain sensor noise.

Resulting graphical causal model

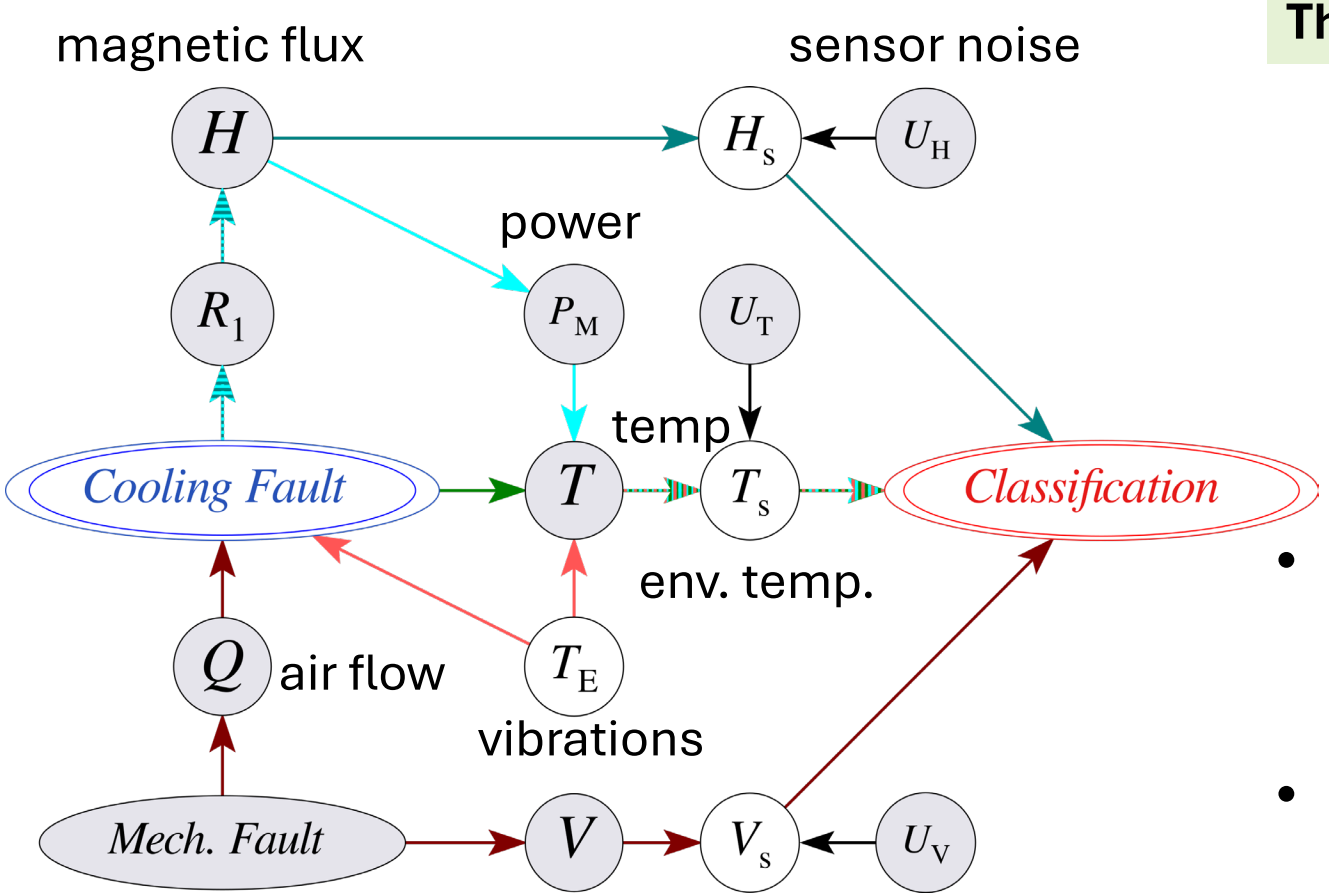


Observations:

- Vibration data alone is insufficient to detect cooling faults, because not all mechanical faults lead to a cooling fault.
- Data on temperature and magnetic flux should be included.
- Sensor data should contain sensor noise.

ID	Requirement (RQ-D: Data Req., RQ-M: Model Req.)
RQ-D1	Training data shall include cases where mechanical faults cause vibrations V without leading to cooling faults .
RQ-D2	The occurrence of cooling faults shall be conditioned on different environmental temperatures T_E such that the model can learn the confounding influence of T_E.
RQ-D3	Measurements shall include characteristic sensor noise.
RQ-M1	Cooling faults shall not be classified based on vibration data V_s alone.
RQ-M2	The input layer shall accept temperature, magnetic flux, and vibration measurements.

Resulting graphical causal model



The causal graph provides testable conditions:

$$\textbf{ID1} : \text{Classification} \perp T_E \mid H_s, T_s, V_s$$

$$\textbf{ID2} : H_s \perp T_E \mid \text{Cooling Fault}$$

$$\textbf{ID3} : H_s \perp V_s \mid \text{Cooling Fault}$$

$$\textbf{ID4} : T_s \perp V_s \mid \text{Cooling Fault}, T_E$$

$$\textbf{ID5} : V_s \perp T_E$$

- ID1: A test case could trigger faults at varying T_E to verify that the detection probability remains unchanged.
- ID5: A monitor could track the correlation between V_S and T_E and trigger an alarm if a threshold is exceeded.

Future work

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Potential avenues for future research:

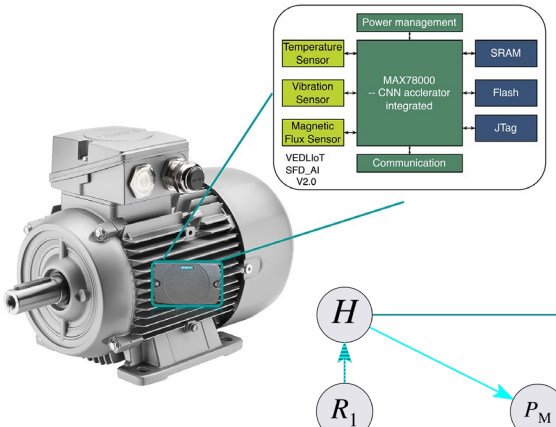
1. How can causal models **complement natural language** requirements? How must causal modelling be adopted to RE?
2. When does a causal graph **include ”enough” prior knowledge**? What is a sufficient set of variables in a causal graph for a given use case?
3. How can the modularization of cause-effect mechanisms help in **modularization of ML software systems**?
4. How can causal models serve as ”**boundary object**” between different stakeholders?
5. To what degree can data requirements from causal models **enhance ML robustness** and **reduce data needs** (more to come)
6. How can **independence conditions** be translated into testing strategies?

Summary

- We suggest to explore causal models and causal reasoning as an RE activity for ML systems.

- In an industrial use case, we showed how causal modelling can be used to formalise prior knowledge.
- We derived data and model requirements, as well as testable conditions from the emerging causal model.

- We proposed a research agenda in which we formulated a set of questions that can lead to establishing causal modelling as an industry practice.



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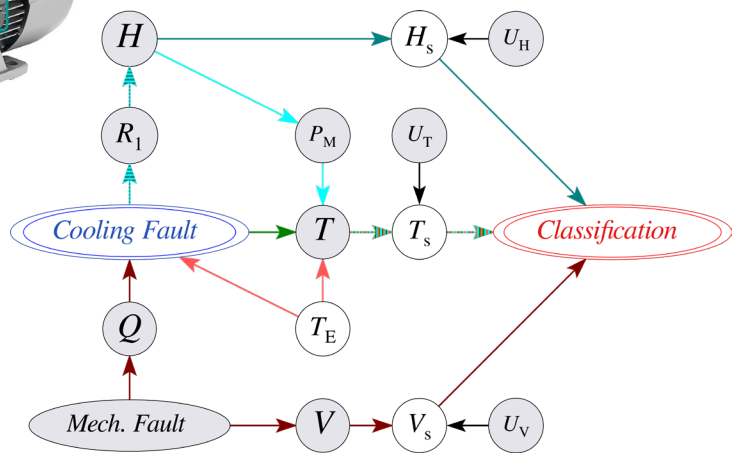

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Conclusion

A Workflow for Causal Inference to take home

Workflow for Statistical Causal Inference

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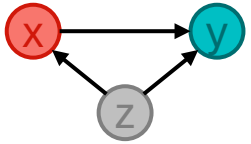
1) Modelling

Creating a DAG of assumed causal relationships



2) Identification

Deriving a statistical from the causal model

Model Derivation

$$y \sim x + z$$

3) Estimation

Training a statistical model with the observed data and evaluating the results

Siebert, J. (2023). Applications of statistical causal inference in software engineering. *Information and Software Technology*, *159*, 107198.

Scientific Workflow: Traditional

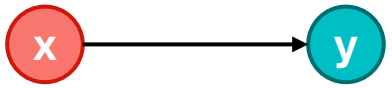
Data Collection

Modelling

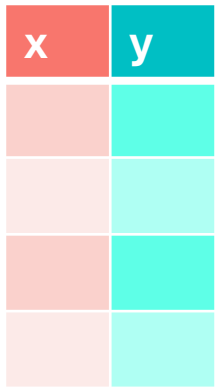
Identification

Estimation

Assumed DAG



Collected Data



Too often, we derive hypotheses from the data available to us.

Scientific Workflow: Better

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Modelling

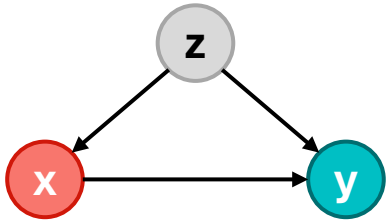
Identification

Data Collection

Estimation

Assumed DAG

hard-to-measure confounder

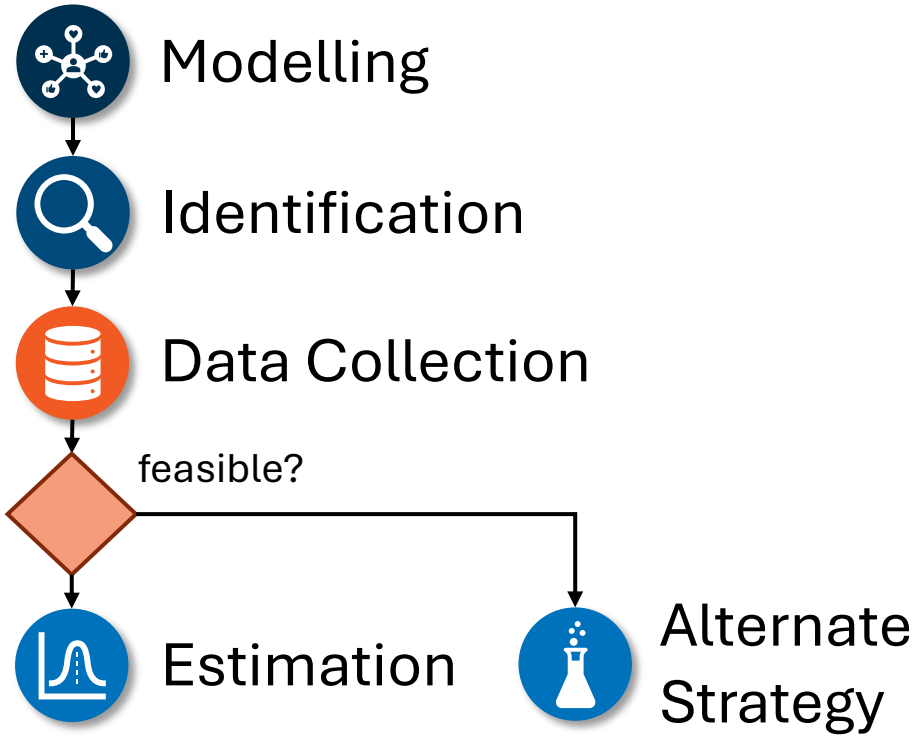


Collected Data

x	y	z

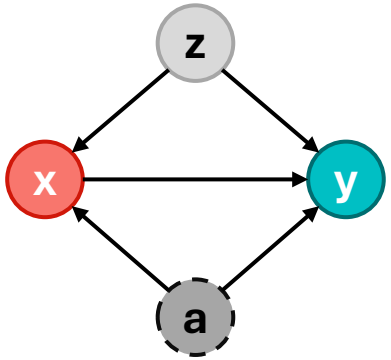
> Determine the variables about which you need to collect data in order to estimate an unbiased effect of interest *before* collecting that data.

Scientific Workflow: Best



Assumed DAG

hard-to-measure confounder



unobservable confounder

Collected Data

x	y	z	a
			x
			x
			x
			x

Unobservable confounders require (a) acknowledgement in the **threats to (internal) validity** and (b) **different research strategies**.

A Vision of Coherent Science

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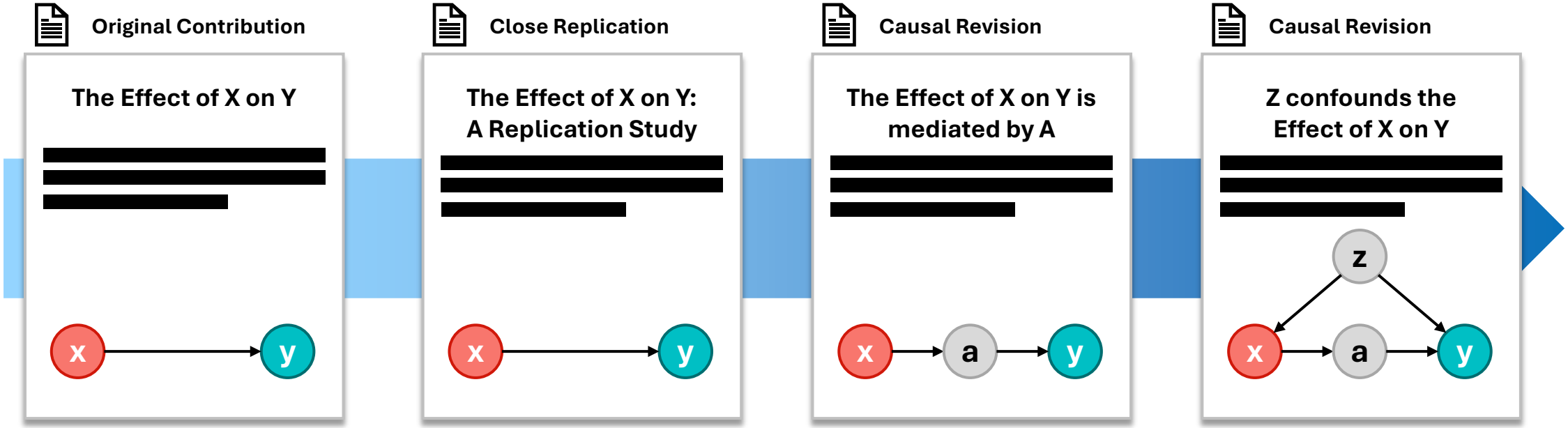
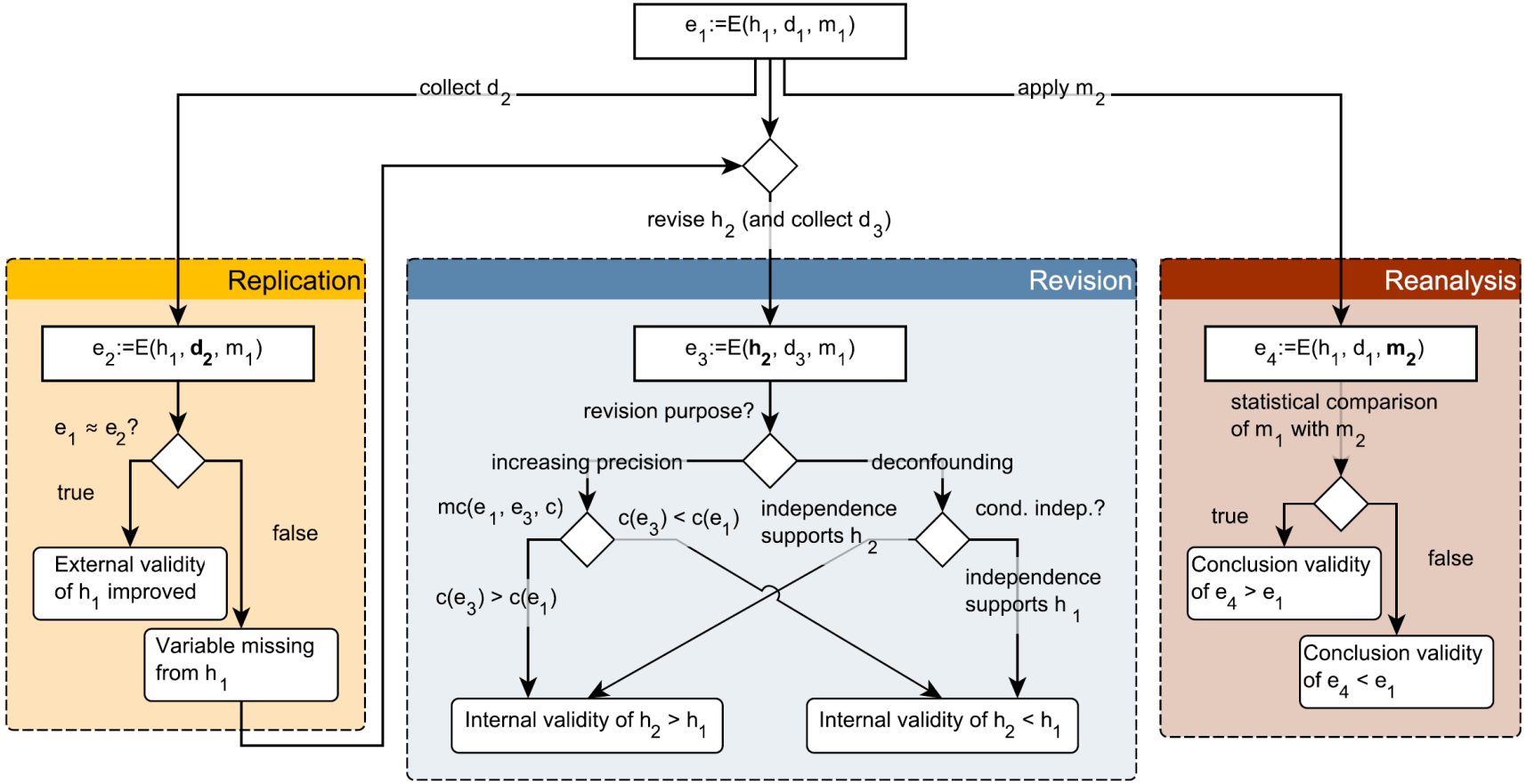
Following this scientific workflow enables us as a community to effectively contribute to a coherent scientific endeavour.



Supporting the Vision of Coherent Science



Frattini, J., Fischbach, J., Fucci, D., Unterkalmsteiner, M., & Mendez, D. (2024). Replications, Revisions, and Reanalyses: Managing Variance Theories in Software Engineering. *arXiv preprint* arXiv:2412.12634.

Closing

Final Remarks

Reading list

Pearl, J., & Mackenzie, D. (2018). *The book of why: the new science of cause and effect*. Basic books.

McElreath, R. (2018). *Statistical rethinking: A Bayesian course with examples in R and Stan*. Chapman and Hall/CRC.

Cinelli, C., **Forney**, A., & **Pearl**, J. (2024). A crash course in good and bad controls. *Sociological Methods & Research*, *53*(3), 1071-1104.

Siebert, J. (2023). Applications of statistical causal inference in software engineering. *Information and Software Technology*, *159*, 107198.

Furia, C. A., **Feldt**, R., & **Torkar**, R. (2019). Bayesian data analysis in empirical software engineering research. *IEEE Transactions on Software Engineering*, *47*(9), 1786-1810.

Furia, C. A., **Torkar**, R., & **Feldt**, R. (2022). Applying Bayesian analysis guidelines to empirical software engineering data: The case of programming languages and code quality. *ACM Transactions on Software Engineering and Methodology (TOSEM)*, *31*(3), 1-38.

Material

All material is public and open-source on GitHub. Feel free to clone, fork, and share the study material.

Comments, requests, and additions are welcome.

JulianFrattini/**bda4sci** Public

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JulianFrattini Reorganized repository structure 9611265 · 5 months ago 23 Commits

version1	Reorganized repository structure	5 months ago
.gitignore	Reorganized repository structure	5 months ago
LICENSE	Initial commit	11 months ago
README.md	Reorganized repository structure	5 months ago
sci-intro.Rproj	Reorganized repository structure	5 months ago

README | Apache-2.0 license

Introduction to Statistical Causal Inference

license Apache-2.0

This repository contains the material for the introduction seminar to Statistical Causal Inference (SCI). The purpose of the seminar is to introduce software engineering researchers with a background in analysis of quantitative data to a causal framework for inferential statistics proposed by Judea Pearl[1] and Richard McElreath[2]

About

Material for the Introduction Seminar to Bayesian Data Analysis for Statistical Causal Inference

bayesian-data-analysis

statistical-causal-inference

Readme

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Activity

4 stars

1 watching

0 forks

Releases 1

Prototype Latest

on Oct 18, 2024

Packages

No packages published

Publish your first package

Limitations

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This was but a gentle introduction. The topic of statistical causal inference contains many more elements, like

- **Ancestors**: parent-child relationships in DAGs
- **Interaction**: moderated inter-relationship between two or more variables and an outcome
- **Hierarchical model**: partial pooling
- **Different types of regression**: regression models for different outcome variable distributions
- **Bayesian data analysis**: advanced estimation technique

And many more …

Ready for more?

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Are you interested in a long-term, in-depth, academic-friendly course about **statistical causal inference** using **Bayesian data analysis**?

→ Contact Julian Frattini and express your interest

We plan to organize a chapter-wise joint reading and discussion of Richard McElreath's "Statistical Rethinking" over the course of a year.

Julian **Frattini**, Ph.D.
Chalmers University of Technology, Sweden
https://julianfrattini.github.io/